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Δ1486-1

FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
CBr4	tetrabromomethane	B	29.5	29.3	0.1	2.5	0.0	1
CCl2O	carbonyl chloride	A	39.5	44.2	-4.8	0.7	0.0	1
CCl2O	carbonyl chloride	A	39.5	44.2	-4.7	0.7	0.0	1
CCl3F	fluorotrichloromethane	A	42.4	40.9	1.5	1.1	0.0	1
CCl3F	fluorotrichloromethane	C	47.8	40.9	6.9	1.1	0.0	2
CCl4	tetrachloromethane	A	30.4	29.3	1.0	2.5	0.0	1
CCl4	tetrachloromethane	C	30.1	29.3	0.8	2.5	0.0	1
* CCl4	tetrachloromethane	A	10.2	29.3	-19.1	2.5	0.0	1
CCl4	tetrachloromethane	B	31.3	29.3	2.0	2.5	0.0	1
CCl4	tetrachloromethane	A	31.3	29.3	2.0	2.5	0.0	2
CF4	tetrafluoromethane	A	27.1	29.3	-2.2	2.5	0.0	2
CF4	tetrafluoromethane	A	30.4	29.3	1.0	2.5	0.0	1
COS	carbonyl sulfide	A	35.2	25.1	10.1	3.0	0.0	1
CS2	carbon disulfide	A	27.2	25.1	2.2	3.0	0.0	1
CSe2	carbon diselenide	B	27.7	25.1	2.6	3.0	0.0	1
CHBr3	tribromomethane	A	39.2	40.9	-1.7	1.1	0.0	2
CHClF2	chlorodifluoromethane	A	36.8	44.2	-7.5	0.7	0.0	1
CHF3	trifluoromethane	A	34.4	40.9	-6.5	1.1	0.0	1
CHN	hydrogen cyanide	A	32.4	30.9	1.6	2.3	0.0	1
(CH2)n	polyethylene	A	9.4	8.7	0.6	0.0	1.0	2
CH2N2	cyanamid	A	22.8	25.1	-2.3	3.0	0.0	2
CH2O2	methanoic acid	A	45.1	50.0	-5.0	0.0	0.0	1
CH2S3	dihydrosulfide carbon sulfide	D	34.2	30.9	3.3	2.3	0.0	1
CH3Br	bromomethane	A	36.0	30.9	5.2	2.3	0.0	1
CH3Cl	chloromethane	A	36.7	30.9	5.8	2.3	0.0	1
CH3Cl2OP	methylphosphonyl dichloride	A	59.1	50.0	9.1	0.0	0.0	1
CH3Cl3Si	trichloromethylsilane	A	45.3	40.9	4.4	1.1	0.0	2
CH3ClFOP	methylphosphonyl chlorofluoride	A	47.3	50.0	-2.7	0.0	0.0	1
CH3DO	methanol-d1	B	21.6	30.9	-9.3	2.3	0.0	1
CH3F2OP	methylphosphonyl difluoride	A	50.3	50.0	0.3	0.0	0.0	1
CH3NO	formamide	A	31.5	50.0	-18.5	0.0	0.0	2
CH3NO	methanamide	B	28.9	50.0	-21.0	0.0	0.0	1
CH3NO2	nitromethane	A	39.6	50.0	-10.3	0.0	0.0	1
CH4	methane	A	14.9	11.7	3.2	4.6	0.0	2
CH4N2O	urea	A	36.4	44.2	-7.9	0.7	0.0	2
CH4N2O	urea	A	34.3	44.2	-10.0	0.7	0.0	2
CH4N2O	urea	B	35.7	44.2	-8.5	0.7	0.0	2
CH4N2O	urea	B	33.5	44.2	-10.7	0.7	0.0	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	CH4O	methanol	B	22.2	30.9	-8.7	2.3	0.0	1
	CH4O	methanol	B	22.5	30.9	-8.3	2.3	0.0	1
*	CH4O	methanol	C	12.5	30.9	-18.3	2.3	0.0	1
	CH4O	methanol	C	21.8	30.9	-9.1	2.3	0.0	1
	CH4O	methanol	A	22.4	30.9	-8.5	2.3	0.0	1
	CH4S	methanethiol	A	40.9	30.9	10.0	2.3	0.0	1
	CH5N	aminomethane	A	34.1	30.9	3.3	2.3	0.0	1
	CH6N2	methylhydrazine	A	47.2	44.2	2.9	0.7	0.0	1
	C2Br2F4	1,2-dibromotetrafluoroethane	A	43.2	44.2	-1.0	0.7	0.0	1
	C2Cl2F4	1,2-dichloro-1,1,2,2-tetrafluoroethane	A	39.0	44.2	-5.3	0.7	0.0	1
	C2Cl3F3	1,1,1-trichlorotrifluoroethane	A	14.3	40.9	-26.5	1.1	0.0	2
	C2Cl3F3	1,1,2-trichloro-1,2,2-trifluoroethane	A	20.5	50.0	-29.5	0.0	0.0	1
*	C2Cl3F3	1,1,2-trichloro-1,2,2-trifluoroethane	A	8.5	50.0	-41.5	0.0	0.0	2
	C2Cl4	tetrachloroethylene	A	48.3	38.5	9.8	1.4	0.0	2
	C2Cl4F2	1,1,1,2-tetrachlorodifluoroethane	A	12.7	50.0	-37.3	0.0	0.0	2
	C2Cl4F2	1,1,2,2-tetrachloro-1,2-difluoroethane	B	12.3	44.2	-31.9	0.7	0.0	1
	C2Cl4F2	1,1,2,2-tetrachloro-1,2-difluoroethane	A	12.3	44.2	-31.9	0.7	0.0	1
	C2Cl6	hexachloroethane	A	53.2	35.1	18.1	1.8	0.0	2
	C2ClF3	chlorotrifluoroethene	A	48.3	50.0	-1.7	0.0	0.0	1
	C2ClF3	chlorotrifluoroethene	A	44.7	50.0	-5.4	0.0	0.0	2
	C2ClF5	pentafluorochloroethane	A	43.6	50.0	-6.4	0.0	0.0	1
	C2F2O2	oxalyl fluoride	A	51.4	44.2	7.2	0.7	0.0	2
	C2F3N	trifluoroacetonitrile	A	38.6	30.9	7.8	2.3	0.0	1
	C2F4	tetrafluoroethene	A	54.3	38.5	15.9	1.4	0.0	1
	C2F4O	trifluoroacetyl fluoride	A	42.8	54.4	-11.5	0.0	0.5	2
	C2F4O	trifluoroacetyl fluoride	A	42.8	54.4	-11.5	0.0	0.5	1
	C2F6	hexafluoroethane	A	51.4	35.1	16.3	1.8	0.0	1
	C2N2	cyanogen	A	33.1	25.1	8.0	3.0	0.0	1
	C2HBrClF3	1-bromo-2-chloro-1,1,2-trifluoroethane	A	30.0	50.0	-20.0	0.0	0.0	2
	C2HBrClF3	2-bromo-2-chloro-1,1,1-trifluoroethane	A	30.8	50.0	-19.2	0.0	0.0	2
	C2HCl3	trichloroethene	A	44.8	50.0	-5.2	0.0	0.0	2
	C2HCl3O2	trichloroacetic acid	D	17.8	54.4	-36.6	0.0	0.5	2
	C2H2Cl2	1,1-dichloroethene	A	43.3	50.0	-6.8	0.0	0.0	1
	C2H2Cl2O2	dichloroacetic acid	D	26.9	54.4	-27.4	0.0	0.5	2
	C2H2Cl2O2	dichloroethanoic acid	B	43.1	54.4	-11.2	0.0	0.5	1
	C2H2Cl4	1,1,2,2-tetrachloroethane	A	43.1	44.2	-1.2	0.7	0.0	2
	C2H2Cl4	1,1,2,2-tetrachloroethane	A	42.4	44.2	-1.9	0.7	0.0	2
	C2H3Cl	vinyl chloride	B	41.2	50.0	-8.8	0.0	0.0	2
	C2H3Cl3	1,1,1-trichloroethane	C	39.7	40.9	-1.2	1.1	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C2H3Cl3	1,1,1-trichloroethane	C	42.2	40.9	1.3	1.1	0.0	1
C2H3Cl3	1,1,1-trichloroethane	A	43.0	40.9	2.1	1.1	0.0	1
C2H3Cl3	1,1,1-trichloroethane	A	41.2	40.9	0.3	1.1	0.0	1
C2H3Cl3	1,1,2-trichloroethane	C	48.0	50.0	-2.0	0.0	0.0	1
C2H3ClF2	1,1-difluoro-1-chloroethane	B	18.9	50.0	-31.1	0.0	0.0	1
C2H3ClF2	1,1-difluoro-1-chloroethane	B	18.9	50.0	-31.1	0.0	0.0	1
C2H3ClO2	monochloroacetic acid	D	48.8	54.4	-5.6	0.0	0.5	2
C2H3F3	1,1,1-trifluoroethane	A	40.2	40.9	-0.7	1.1	0.0	1
C2H3N	acetonitrile	A	39.8	30.9	8.9	2.3	0.0	1
C2H3N	acetonitrile	C	32.8	30.9	2.0	2.3	0.0	2
C2H4	ethylene	B	32.2	25.1	7.1	3.0	0.0	2
C2H4	ethylene	A	32.2	25.1	7.1	3.0	0.0	2
C2H4Br2	1,2-dibromoethane	A	45.8	44.2	1.5	0.7	0.0	1
C2H4Br2	1,2-dibromoethane	C	46.5	44.2	2.2	0.7	0.0	1
C2H4Br4Cl	1-bromo-2-chloroethane	C	54.6	50.0	4.6	0.0	0.0	1
C2H4Cl2	1,1-dichloroethane	A	44.7	50.0	-5.3	0.0	0.0	1
C2H4Cl2	1,2-dichloroethane	A	37.3	44.2	-7.0	0.7	0.0	1
C2H4Cl2	1,2-dichloroethane	C	53.1	44.2	8.8	0.7	0.0	1
C2H4D2O2	1,2-dihydroxyethane-d2	B	37.7	44.2	-6.6	0.7	0.0	1
C2H4O	ethanal	A	22.7	50.0	-27.3	0.0	0.0	2
C2H4O	oxirane	A	32.2	44.2	-12.0	0.7	0.0	1
C2H4O2	acetic acid	D	38.4	50.0	-11.6	0.0	0.0	2
C2H4O2	acetic acid (ethanoic acid)	A	39.2	50.0	-10.7	0.0	0.0	1
C2H4O2	acetic acid (ethanoic acid)	C	40.5	50.0	-9.5	0.0	0.0	1
C2H5Cl	chloroethane	A	33.0	50.0	-16.9	0.0	0.0	1
C2H5DO	ethanol-d1	A	57.0	50.0	7.0	0.0	0.0	1
C2H5NO	acetamide	A	44.1	50.0	-5.9	0.0	0.0	2
C2H5NO	acetamide	A	44.2	50.0	-5.9	0.0	0.0	2
C2H5NO	acetamide	A	37.1	50.0	-12.8	0.0	0.0	2
C2H5NO2	methyl carbamate	A	50.8	50.0	0.8	0.0	0.0	2
C2H5NO2	nitroethane	A	53.6	50.0	3.6	0.0	0.0	1
C2H5OS	dimethyl sulfoxide	A	49.3	44.2	5.0	0.7	0.0	1
(C2H6OSi)n	poly(dimethylsiloxane)	A	18.5	8.7	9.8	0.0	1.0	2
C2H6	ethane	A	31.2	25.1	6.1	3.0	0.0	1
C2H6	ethane	A	31.9	25.1	6.8	3.0	0.0	1
C2H6	ethane	A	31.8	25.1	6.7	3.0	0.0	2
C2H6Cd	dimethyl cadmium	A	35.0	44.2	-9.3	0.7	0.0	1
C2H6Cl2Si	dichlorodimethylsilane	A	44.4	50.0	-5.6	0.0	0.0	2
C2H6N2O	methylurea	A	42.1	54.4	-12.2	0.0	0.5	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C2H6O	2-oxapropane	A	37.5	44.2	-6.8	0.7	0.0	1
*	C2H6O	ethanol	B	31.3	50.0	-18.7	0.0	0.0	2
	C2H6O	ethanol	A	57.4	50.0	7.4	0.0	0.0	1
	C2H6O	ethanol	A	36.2	50.0	-13.8	0.0	0.0	1
*	C2H6O	ethanol	C	31.3	50.0	-18.7	0.0	0.0	1
*	C2H6O	ethanol	B	31.7	50.0	-18.3	0.0	0.0	1
	C2H6O2	1,2-dihydroxyethane	C	44.6	44.2	0.3	0.7	0.0	1
	C2H6O2	1,2-dihydroxyethane	B	38.2	44.2	-6.0	0.7	0.0	1
	C2H6O2S	dimethyl sulfone	A	47.9	44.2	3.7	0.7	0.0	1
	C2H6S	dimethylsulfide	A	45.7	44.2	1.4	0.7	0.0	1
+	C2H6S	ethanethiol	A	39.7	50.0	-10.2	0.0	0.0	1
	C2H6S2	2,3-dithiabutane	A	48.8	58.7	-9.9	0.0	1.0	1
	C2H7N	dimethylamine	A	32.8	44.2	-11.4	0.7	0.0	1
	C2H8N2	1,2-diaminoethane	A	82.0	44.2	37.8	0.7	0.0	1
	C2H8N2	n,n'-dimethylhydrazine	A	51.6	44.2	7.4	0.7	0.0	2
	C2H8N2	n,n-dimethylhydrazine	A	46.6	44.2	2.4	0.7	0.0	1
	C2H12B10	m-carborane	A	27.7	44.2	-16.5	0.7	0.0	2
	C3F6O	hexafluoropropanone	A	56.8	44.2	12.5	0.7	0.0	1
	C3F8	perfluoropropane	A	39.6	44.2	-4.7	0.7	0.0	1
	C3H2Cl3F3	1,1,1-trichloro-3,3,3-trifluoropropane	A	60.5	67.4	-7.0	0.0	2.1	1
	C3H2ClF5	3-chloro-1,1,1,3,3-pentafluoropropane	A	63.3	67.4	-4.1	0.0	2.1	1
	C3H2N2	malononitrile	A	35.4	44.2	-8.9	0.7	0.0	1
	C3H3Cl2F3	1,1,1-trifluoro-3,3-dichloropropane	A	51.1	67.4	-16.3	0.0	2.1	1
	C3H3Cl3	1,1,2-trichloroethane	A	45.7	50.0	-4.3	0.0	0.0	2
	C3H3N	acrylonitrile	A	40.2	50.0	-9.8	0.0	0.0	1
	C3H3N3	s-triazine	B	41.2	35.1	6.1	1.8	0.0	1
	C3H3NS	thiazole	A	39.8	50.0	-10.1	0.0	0.0	1
	C3H3NS	thiazole	A	40.0	50.0	-10.0	0.0	0.0	1
	C3H4Cl3Si	beta-trichlorosilylpropionitrile	A	69.0	67.4	1.6	0.0	2.1	2
	C3H4Cl4	1,1,1,3-tetrachloropropane	A	54.2	67.4	-13.2	0.0	2.1	1
	C3H4ClF3	1,1,1-trifluoro-3-chloropropane	A	56.1	67.4	-11.3	0.0	2.1	1
	C3H4ClF3	1,1,1-trifluoro-3-chloropropane	A	54.6	67.4	-12.8	0.0	2.1	1
	C3H4N2	imidazole	A	35.4	50.0	-14.6	0.0	0.0	2
	C3H4N2	imidazole	B	35.4	50.0	-14.6	0.0	0.0	2
	C3H4N2	pyrazole	B	41.4	50.0	-8.6	0.0	0.0	2
	C3H4N2O	cyanoacetamide	B	59.5	54.4	5.1	0.0	0.5	2
	C3H4O2	acrylic acid	A	33.3	50.0	-16.7	0.0	0.0	2
	C3H4O2	beta-propiolactone	A	39.2	50.0	-10.7	0.0	0.0	2
	C3H4O3	ethylene carbonate	B	43.0	44.2	-1.3	0.7	0.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C3H5N	propionitrile	A	37.5	50.0	-12.4	0.0	0.0	1
(C3H6O)n	poly(oxacyclobutane)	B	31.0	34.8	-3.8	0.0	4.2	1
(C3H6O2)n	poly-1,3-dioxolan	B	51.4	43.5	7.9	0.0	5.2	2
C3H6	cyclopropane	A	37.4	35.1	2.3	1.8	0.0	1
C3H6	propene	B	33.3	50.0	-16.7	0.0	0.0	1
C3H6	propylene	A	34.2	50.0	-15.8	0.0	0.0	2
C3H6	propylene	B	34.2	50.0	-15.8	0.0	0.0	2
C3H6Br2	1,3-dibromopropane	C	61.4	67.4	-6.1	0.0	2.1	1
C3H6Cl2	2,2-dichloropropane	B	41.6	44.2	-2.7	0.7	0.0	1
C3H6ClNO2	2-chloro-2-nitropropane	C	50.0	50.0	-0.1	0.0	0.0	1
C3H6N2O4	2,2-dinitropropane	B	57.4	58.7	-1.3	0.0	1.0	2
C3H6N2O4	2,2-dinitropropane	B	57.5	58.7	-1.3	0.0	1.0	1
C3H6O	acetone	A	32.4	44.2	-11.8	0.7	0.0	1
C3H6O	propanal	A	50.2	54.4	-4.2	0.0	0.5	2
C3H6O	propanone	C	32.0	44.2	-12.2	0.7	0.0	1
C3H6O	propanone	C	32.0	44.2	-12.2	0.7	0.0	1
C3H6O	propanone	C	26.7	44.2	-17.5	0.7	0.0	1
C3H6O	propylene oxide	C	40.7	50.0	-9.3	0.0	0.0	1
C3H6O	propylene oxide	A	40.5	50.0	-9.5	0.0	0.0	1
C3H6O2	1,3-dioxolan	B	56.1	50.0	6.1	0.0	0.0	2
C3H6O2	propanoic acid	A	42.2	54.4	-12.1	0.0	0.5	1
C3H6O2S	3-thiolpropanoic acid	C	58.2	63.1	-4.9	0.0	1.6	1
C3H6O3	2-hydroxypropanoic acid (dl)	C	39.1	54.4	-15.2	0.0	0.5	1
C3H6O3	2-hydroxypropanoic acid (dl)	B	39.1	54.4	-15.2	0.0	0.5	2
C3H6S	thiacyclobutane	A	45.1	44.2	0.8	0.7	0.0	1
C3H7Br	2-bromopropane	B	35.5	44.2	-8.8	0.7	0.0	1
C3H7N	cyclopropylamine	A	55.4	44.2	11.2	0.7	0.0	1
C3H7NO2	ethyl carbamate	A	52.3	54.4	-2.1	0.0	0.5	2
C3H7NO2	ethyl carbamate	A	65.0	54.4	10.6	0.0	0.5	2
C3H8	propane	A	41.2	44.2	-3.0	0.7	0.0	1
C3H8N2O	1,1-dimethylurea	A	65.2	54.4	10.9	0.0	0.5	2
C3H8N2O	1,3-dimethylurea	A	35.9	63.1	-27.1	0.0	1.6	2
C3H8N2O	ethylurea	A	37.9	63.1	-25.1	0.0	1.6	2
C3H8O	1-propanol	C	35.3	50.0	-14.6	0.0	0.0	1
C3H8O	1-propanol	A	36.1	50.0	-13.9	0.0	0.0	1
C3H8O	2-propanol	C	28.7	44.2	-15.5	0.7	0.0	1
C3H8O	2-propanol	C	28.7	44.2	-15.5	0.7	0.0	1
C3H8O	2-propanol	A	29.1	44.2	-15.1	0.7	0.0	1
C3H8O	2-propanol	A	29.2	44.2	-15.0	0.7	0.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C3H8O2	methylal	A	49.6	67.4	-17.8	0.0	2.1	1
C3H8O3	1,2,3-trihydroxypropane	B	62.7	67.4	-4.7	0.0	2.1	1
C3H8O3	1,2,3-trihydroxypropane	B	62.8	67.4	-4.6	0.0	2.1	2
C3H8S	1-propanethiol	A	62.2	50.0	12.2	0.0	0.0	1
C3H8S	2-propanethiol	A	40.7	44.2	-3.6	0.7	0.0	1
C3H8S	thiabutane	A	58.4	58.7	-0.3	0.0	1.0	1
C3H9Al	trimethylaluminum	A	30.5	40.9	-10.3	1.1	0.0	2
C3H9As	trimethyl arsine	A	48.0	40.9	7.2	1.1	0.0	2
C3H9B	trimethylborane	A	28.7	40.9	-12.1	1.1	0.0	1
C3H9ClSi	chlorotrimethylsilane	B	48.2	40.9	7.3	1.1	0.0	2
C3H9Ga	trimethylgallium	B	42.8	40.9	2.0	1.1	0.0	1
C3H9Ga	trimethylgallium	A	42.5	40.9	1.6	1.1	0.0	2
C3H9N	1-aminopropane	B	56.4	58.7	-2.3	0.0	1.0	1
C3H9N	1-aminopropane	A	58.3	58.7	-0.4	0.0	1.0	1
C3H9N	2-aminopropane	A	41.2	44.2	-3.1	0.7	0.0	1
C3H9N	trimethylamine	A	41.9	40.9	1.1	1.1	0.0	1
C3H10N2	1,2-diaminopropane	A	78.2	58.7	19.5	0.0	1.0	1
C3H10N2	trimethylhydrazine	A	47.1	58.7	-11.5	0.0	1.0	1
C3H12BN	trimethylamineborane	A	22.3	40.9	-18.5	1.1	0.0	1
C4F8	octafluorocyclobutane	A	11.9	32.7	-20.8	2.1	0.0	1
C4H2O3	maleic anhydride	D	37.6	44.2	-6.6	0.7	0.0	1
C4H2O3	maleic anhydride	C	39.8	44.2	-4.5	0.7	0.0	1
C4H2O3	maleic anhydride	B	41.8	44.2	-2.4	0.7	0.0	2
C4H2O3	maleic anhydride	A	41.6	44.2	-2.6	0.7	0.0	2
C4H3Cl3OS	methyl trichlorothioacrylate	A	71.2	63.1	8.1	0.0	1.6	2
C4H3F5O3	alpha-(trifluoromethoxy)-alpha,alpha-difluoromethyl acetate	A	50.8	89.2	-38.3	0.0	4.7	2
(C4H4O4)n	polyglycolide	A	46.9	43.5	3.4	0.0	5.2	2
C4H4N2	succinonitrile	B	11.1	38.5	-27.3	1.4	0.0	2
C4H4N2	succinonitrile (1,4-butanedinitrile)	A	37.8	38.5	-0.7	1.4	0.0	1
C4H4O	furan	A	33.9	44.2	-10.3	0.7	0.0	1
C4H4O4	ethylene oxalate	A	32.3	44.2	-11.9	0.7	0.0	2
C4H4O4	glycolide	B	47.4	44.2	3.1	0.7	0.0	1
C4H4O4	glycolide	A	47.5	44.2	3.2	0.7	0.0	2
C4H4O4	glycolide	A	47.5	44.2	3.2	0.7	0.0	2
C4H4O4	glycolide	A	47.4	44.2	3.1	0.7	0.0	2
C4H4O4	glycolide	A	47.4	44.2	3.1	0.7	0.0	2
C4H4S	thiophene	A	25.4	44.2	-18.8	0.7	0.0	1
C4H4S	thiophene	C	28.3	44.2	-15.9	0.7	0.0	1
C4H5ClO2	cis-3-chloro-2-butenoic acid	C	41.4	54.4	-12.9	0.0	0.5	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C4H5ClO2	trans-3-chloro-2-butenoic acid	C	56.5	54.4	2.1	0.0	0.5	1
C4H5N	pyrrole	A	31.7	44.2	-12.5	0.7	0.0	1
C4H5NS	2-methylthiazole	A	49.0	50.0	-1.1	0.0	0.0	1
C4H5NS	2-methylthiazole	A	49.0	50.0	-1.0	0.0	0.0	1
C4H5NS	2-methylthiazole	A	49.0	50.0	-1.0	0.0	0.0	1
(C4H6)n	cis-1,4-polybutadiene	B	15.1	17.4	-2.3	0.0	2.1	1
C4H6	1,2-butadiene	A	50.9	50.0	0.9	0.0	0.0	1
C4H6	1,3-butadiene	A	48.6	44.2	4.4	0.7	0.0	1
C4H6	1-butyne	A	40.9	50.0	-9.1	0.0	0.0	1
C4H6	2-butyne	A	38.3	25.1	13.3	3.0	0.0	1
C4H6O2	gamma-butyrolactone	A	41.6	50.0	-8.4	0.0	0.0	2
C4H6O2	methacrylic acid	A	28.0	54.4	-26.3	0.0	0.5	2
C4H6O2	methyl propenoate	A	49.3	58.7	-9.5	0.0	1.0	2
C4H6O3	propylene carbonate	A	42.8	63.1	-20.2	0.0	1.6	2
C4H6O3	propylene carbonate	A	42.8	63.1	-20.3	0.0	1.6	2
C4H7NO	alpha-pyrrolidone	B	46.5	50.0	-3.5	0.0	0.0	2
(C4H8O)n	polytetrahydrofuran	A	34.8	43.5	-8.7	0.0	5.2	2
(C4H8O)n	polytetrahydrofuran	A	34.8	43.5	-8.7	0.0	5.2	2
C4H8	1-butene	A	43.8	50.0	-6.2	0.0	0.0	2
C4H8	1-butene	A	43.8	50.0	-6.2	0.0	0.0	1
C4H8	2-methylpropene	C	44.7	44.2	0.5	0.7	0.0	1
C4H8	cis-2-butene	A	54.4	44.2	10.2	0.7	0.0	1
C4H8	cis-2-butene	A	54.4	44.2	10.2	0.7	0.0	2
C4H8	cis-2-butene	C	54.7	44.2	10.5	0.7	0.0	1
C4H8	cyclobutane	A	45.1	32.7	12.4	2.1	0.0	1
C4H8	trans-2-butene	C	58.9	44.2	14.7	0.7	0.0	1
C4H8	trans-2-butene	A	58.2	44.2	14.0	0.7	0.0	1
C4H8	trans-2-butene	A	58.2	44.2	14.0	0.7	0.0	2
C4H8O	butanal	C	62.8	63.1	-0.3	0.0	1.6	1
C4H8O	butanone	A	45.3	54.4	-9.1	0.0	0.5	1
C4H8O	butanone	A	45.0	54.4	-9.4	0.0	0.5	1
C4H8O	butanone	C	45.6	54.4	-8.8	0.0	0.5	1
C4H8O	tetrahydrofuran	A	51.8	44.2	7.6	0.7	0.0	1
C4H8O	tetrahydrofuran	A	51.8	44.2	7.6	0.7	0.0	1
C4H8O	tetrahydrofuran	A	51.8	44.2	7.6	0.7	0.0	2
C4H8O	tetrahydrofuran	A	51.8	44.2	7.6	0.7	0.0	2
C4H8O2	1,4-dioxane	C	53.8	38.5	15.4	1.4	0.0	1
C4H8O2	1,4-dioxane	C	42.0	38.5	3.5	1.4	0.0	1
C4H8O2	butanoic acid	A	48.7	63.1	-14.4	0.0	1.6	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C4H8O2	butanoic acid	C	41.4	63.1	-21.6	0.0	1.6	1
C4H8O2	ethyl ethanoate	C	55.4	63.1	-7.7	0.0	1.6	1
C4H8O4	tetroxan	C	58.7	50.0	8.7	0.0	0.0	1
C4H8S	thiacyclopentane	A	41.5	44.2	-2.7	0.7	0.0	1
C4H8S2	1,3-dithiane	B	46.5	50.0	-3.5	0.0	0.0	2
C4H8S2	1,4-dithiane	B	56.2	44.2	11.9	0.7	0.0	2
C4H9Br	1-bromobutane	C	57.6	67.4	-9.8	0.0	2.1	1
C4H9Br	2-bromo-2-methylpropane	B	39.3	40.9	-1.6	1.1	0.0	1
C4H9Cl	2-chloro-2-methylpropane	A	45.4	40.9	4.6	1.1	0.0	1
C4H9Cl	2-chloro-2-methylpropane	B	44.0	40.9	3.1	1.1	0.0	1
C4H9N	pyrrolidine	A	42.5	44.2	-1.8	0.7	0.0	1
C4H9N	pyrrolidine	A	42.5	44.2	-1.8	0.7	0.0	1
(C4H10OSi)n	poly(diethylsiloxane)	A	20.0	26.1	-6.2	0.0	3.1	2
(C4H10OSi)n	poly(diethylsiloxane)	A	19.9	26.1	-6.2	0.0	3.1	2
C4H10	2-methylpropane	C	39.7	40.9	-1.1	1.1	0.0	1
C4H10	2-methylpropane	A	39.9	40.9	-0.9	1.1	0.0	1
C4H10	n-butane	C	52.4	58.7	-6.3	0.0	1.0	1
C4H10	n-butane	A	53.8	58.7	-5.0	0.0	1.0	1
C4H10Hg	diethyl mercury	A	57.9	67.4	-9.5	0.0	2.1	2
C4H10N2O	1,1,3-trimethylurea	A	41.5	63.1	-21.5	0.0	1.6	2
C4H10N2O	isopropylurea	A	46.9	63.1	-16.2	0.0	1.6	2
C4H10N2O	propylurea	A	38.4	71.8	-33.3	0.0	2.6	2
C4H10O	1-butanol	C	50.5	67.4	-16.9	0.0	2.1	1
C4H10O	1-butanol	A	50.8	67.4	-16.6	0.0	2.1	1
C4H10O	2-butanol	A	33.8	58.7	-24.8	0.0	1.0	1
C4H10O	2-butanol	A	32.3	58.7	-26.3	0.0	1.0	1
C4H10O	2-methyl-1-propanol	A	36.9	58.7	-21.7	0.0	1.0	1
C4H10O	2-methyl-2-propanol	C	22.7	40.9	-18.1	1.1	0.0	1
C4H10O	2-methyl-2-propanol	A	27.0	40.9	-13.8	1.1	0.0	1
C4H10O	2-oxa-3-methylbutane	A	45.7	58.7	-12.9	0.0	1.0	1
C4H10O	2-oxapentane	A	57.2	67.4	-10.1	0.0	2.1	1
C4H10O	3-oxapentane	C	46.6	67.4	-20.8	0.0	2.1	1
C4H10O	3-oxapentane	A	45.8	67.4	-21.5	0.0	2.1	1
C4H10O2	1,4-butanediol	A	63.7	76.1	-12.4	0.0	3.1	2
* C4H10O2Se	beta-selenodiglycol	A	0.7	84.8	-84.1	0.0	4.2	2
C4H10O3	1,1,1-trihydroxymethylpropane	A	64.3	58.7	5.6	0.0	1.0	2
C4H10O4	1,2,3,4-tetrahydroxybutane	B	111.0	76.1	34.9	0.0	3.1	1
C4H10S	1-butanethiol	A	66.4	67.4	-1.0	0.0	2.1	1
C4H10S	2-butanethiol	A	48.7	58.7	-10.0	0.0	1.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C4H10S	2-methyl-1-propanethiol	A	38.8	58.7	-19.8	0.0	1.0	1
C4H10S	2-methyl-2-propanethiol	A	44.9	40.9	4.0	1.1	0.0	1
C4H10S	2-thiapentane	A	61.9	67.4	-5.5	0.0	2.1	1
C4H10S	3-methyl-2-thiabutane	A	54.5	58.7	-4.2	0.0	1.0	1
C4H10S	3-thiapentane	A	70.4	67.4	2.9	0.0	2.1	1
C4H10S2	diethyl disulfide	A	54.8	76.1	-21.3	0.0	3.1	1
C4H11N	2-methyl-2-aminopropane	A	35.4	40.9	-5.4	1.1	0.0	1
C4H12Ge	tetramethylgermane	A	40.4	29.3	11.1	2.5	0.0	2
C4H12N2	1,2-diamino-2-methylpropane	A	73.8	58.7	15.1	0.0	1.0	1
C4H12Pb	tetramethyl lead	B	44.4	29.3	15.1	2.5	0.0	2
C4H12Si	tetramethylsilane	A	38.7	29.3	9.4	2.5	0.0	1
C4H12Si	tetramethylsilane	A	34.9	29.3	5.6	2.5	0.0	1
C4H12Si	tetramethylsilane	A	39.5	29.3	10.1	2.5	0.0	1
C4H12Sn	tetramethyl tin	A	42.4	29.3	13.0	2.5	0.0	2
C4H12Sn	tetramethyl tin	B	43.3	29.3	13.9	2.5	0.0	2
C5F11N	perfluoropiperidine	A	62.1	50.0	12.1	0.0	0.0	1
C5F13N	perfluoromethyldiethylamine	A	30.7	67.4	-36.7	0.0	2.1	2
C5F13N	perfluoromethyldiethylamine	A	91.8	67.4	24.3	0.0	2.1	1
C5H3F7O2	methyl perfluorobutanoate	A	61.5	80.5	-18.9	0.0	3.7	2
C5H4O2	furfural	C	61.1	50.0	11.1	0.0	0.0	1
C5H5F3O2	trifluoromethyl(2-hydroxy-1-propenyl) ketone	A	36.4	67.4	-31.0	0.0	2.1	2
* C5H5N	pyridine	C	13.5	44.2	-30.7	0.7	0.0	1
C5H5N	pyridine	C	35.8	44.2	-8.5	0.7	0.0	1
C5H5N	pyridine	A	35.8	44.2	-8.5	0.7	0.0	1
C5H6	cyclopentadiene	A	45.4	44.2	1.1	0.7	0.0	2
C5H6N2	dimethylmalonitrile	A	45.8	44.2	1.6	0.7	0.0	1
C5H6N2	glutaronitrile	A	51.5	67.4	-15.8	0.0	2.1	1
C5H6N2O2	thymine	C	54.5	50.0	4.5	0.0	0.0	2
C5H6O	2-methylfuran	A	47.0	50.0	-3.0	0.0	0.0	1
C5H6O2	furfuryl alcohol	C	58.4	54.4	4.1	0.0	0.5	1
C5H6O2	furfuryl alcohol	C	50.8	54.4	-3.6	0.0	0.5	1
C5H6S	2-methylthiophene	A	53.6	50.0	3.6	0.0	0.0	1
C5H7N	n-methylpyrrole	A	36.1	44.2	-8.2	0.7	0.0	2
C5H7N	n-methylpyrrole	A	36.1	44.2	-8.2	0.7	0.0	2
C5H7NO2	ethyl cyanoacetate	A	47.7	71.8	-24.0	0.0	2.6	2
(C5H8)n	cis-polypentenamer	B	22.4	34.8	-12.4	0.0	4.2	2
(C5H8)n	polypentenamer	A	27.6	34.8	-7.3	0.0	4.2	2
(C5H8)n	polypentenamer	A	27.6	34.8	-7.3	0.0	4.2	2
(C5H8)n	trans-polypentenamer	B	28.7	34.8	-6.1	0.0	4.2	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C5H8	1,2-pentadiene	A	55.6	54.4	1.3	0.0	0.5	1
C5H8	1,4-pentadiene	C	49.4	58.7	-9.3	0.0	1.0	1
C5H8	1,4-pentadiene	A	48.6	58.7	-10.0	0.0	1.0	1
C5H8	1-cis-3-pentadiene	A	42.6	54.4	-11.7	0.0	0.5	1
C5H8	1-trans-3-pentadiene	A	38.5	54.4	-15.8	0.0	0.5	1
C5H8	2,3-pentadiene	A	41.6	44.2	-2.7	0.7	0.0	1
C5H8	2-methyl-1,3-butadiene	A	38.7	50.0	-11.3	0.0	0.0	1
C5H8	2-methyl-1,3-butadiene	A	38.2	50.0	-11.7	0.0	0.0	1
C5H8	3-methyl-1,2-butadiene	A	49.9	44.2	5.6	0.7	0.0	1
C5H8	cyclopentene	A	29.9	44.2	-14.3	0.7	0.0	1
C5H8	methylenecyclobutane	A	41.5	44.2	-2.7	0.7	0.0	2
C5H8	methylenecyclobutane	A	41.5	44.2	-2.7	0.7	0.0	2
C5H8	methylenecyclobutane	A	42.3	44.2	-1.9	0.7	0.0	2
C5H8	methylenecyclobutane	A	41.5	44.2	-2.7	0.7	0.0	1
C5H8	spiropentane	A	38.7	38.5	0.3	1.4	0.0	1
C5H8Br4	2,2-bis(bromomethyl)-1,3-dibromopropane	A	64.5	84.8	-20.3	0.0	4.2	1
C5H8Cl2O	3,3-bis-(chloromethyl)oxacyclobutane	A	58.0	63.1	-5.1	0.0	1.6	1
C5H8F4	2,2-bis(fluoromethyl)-1,3-difluoropropane	A	66.9	84.8	-17.9	0.0	4.2	1
C5H8O2	acetylacetone	B	56.9	58.7	-1.8	0.0	1.0	2
C5H8O2	delta-valerolactone	A	47.2	50.0	-2.8	0.0	0.0	2
C5H8O2	methyl 2-methylpropenoate	A	59.7	54.4	5.3	0.0	0.5	2
C5H8O2	methyl 2-methylpropenoate	B	64.0	54.4	9.6	0.0	0.5	2
C5H9N	2,2-dimethylpropionitrile	A	41.1	40.9	0.2	1.1	0.0	1
C5H9NO	alpha-piperidone	B	47.0	50.0	-3.0	0.0	0.0	2
C5H10	1-pentene	A	53.8	63.1	-9.2	0.0	1.6	1
C5H10	1-pentene	A	53.8	63.1	-9.2	0.0	1.6	2
C5H10	2-methyl-1-butene	A	58.3	54.4	4.0	0.0	0.5	2
C5H10	2-methyl-1-butene	A	58.3	54.4	4.0	0.0	0.5	1
C5H10	2-methyl-2-butene	C	53.5	50.0	3.5	0.0	0.0	1
C5H10	2-methyl-2-butene	A	54.4	50.0	4.4	0.0	0.0	2
C5H10	2-methyl-2-butene	A	54.5	50.0	4.5	0.0	0.0	1
C5H10	3-methyl-1-butene	A	51.2	54.4	-3.2	0.0	0.5	2
C5H10	3-methyl-1-butene	A	51.2	54.4	-3.2	0.0	0.5	1
C5H10	cis-2-pentene	A	58.4	58.7	-0.3	0.0	1.0	1
C5H10	cis-2-pentene	A	58.4	58.7	-0.3	0.0	1.0	2
C5H10	cyclopentane	A	45.8	30.9	14.9	2.3	0.0	1
C5H10	cyclopentane	A	45.7	30.9	14.8	2.3	0.0	1
C5H10	cyclopentane	A	45.8	30.9	14.9	2.3	0.0	1
C5H10	cyclopentane	C	45.0	30.9	14.2	2.3	0.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C5H10	trans-2-pentene	A	62.8	58.7	4.1	0.0	1.0	1
C5H10	trans-2-pentene	A	62.8	58.7	4.1	0.0	1.0	2
C5H10O	2,2-dimethylpropanal	A	38.6	54.4	-15.7	0.0	0.5	2
C5H10O	2-pentanone	A	54.1	63.1	-9.0	0.0	1.6	1
C5H10O	2-pentanone	A	56.3	63.1	-6.7	0.0	1.6	1
C5H10O	3-methylbutanone	A	51.9	54.4	-2.4	0.0	0.5	1
C5H10O	3-pentanone	A	50.5	63.1	-12.5	0.0	1.6	1
C5H10O	cyclopentanol	C	24.3	44.2	-19.9	0.7	0.0	1
C5H10O2	pentanoic acid	A	59.1	71.8	-12.6	0.0	2.6	1
C5H10O5	pentoxan	C	65.6	50.0	15.6	0.0	0.0	1
C5H10S	2-methylthiolane	A	51.5	50.0	1.5	0.0	0.0	1
C5H10S	3-methylthiolane	A	54.0	50.0	4.0	0.0	0.0	1
C5H10S	cyclopentanethiol	A	50.4	50.0	0.4	0.0	0.0	1
C5H10S	thiacyclohexane	A	46.2	50.0	-3.8	0.0	0.0	1
C5H11Br	1-bromopentane	B	61.9	76.1	-14.1	0.0	3.1	1
C5H11Br	1-bromopentane	C	77.6	76.1	1.5	0.0	3.1	1
C5H11N	cyclopentylamine	A	46.2	44.2	2.0	0.7	0.0	1
C5H11N	piperidine	A	56.7	44.2	12.4	0.7	0.0	2
C5H11N	piperidine	A	56.7	44.2	12.4	0.7	0.0	2
C5H12	2,2-dimethylpropane	A	31.1	29.3	1.8	2.5	0.0	1
C5H12	2,2-dimethylpropane	A	30.7	29.3	1.4	2.5	0.0	1
C5H12	2-methylbutane	A	45.2	58.7	-13.4	0.0	1.0	1
C5H12	2-methylbutane	C	45.4	58.7	-13.3	0.0	1.0	1
C5H12	2-methylbutane	A	45.5	58.7	-13.2	0.0	1.0	1
C5H12	n-pentane	C	58.4	67.4	-9.0	0.0	2.1	1
C5H12	n-pentane	A	58.5	67.4	-8.9	0.0	2.1	1
C5H12	n-pentane	A	58.6	67.4	-8.8	0.0	2.1	1
C5H12N2O	1,3-diethylurea	A	87.6	80.5	7.1	0.0	3.7	2
C5H12N2O	butylurea	A	64.4	80.5	-16.1	0.0	3.7	2
C5H12N2O	n,n-diethylurea	B	51.4	71.8	-20.3	0.0	2.6	2
C5H12N2O	n,n-diethylurea	A	49.0	71.8	-22.7	0.0	2.6	2
C5H12N2O	tert-butylurea	A	73.7	54.4	19.3	0.0	0.5	2
C5H12O	1-pentanol	A	53.7	76.1	-22.4	0.0	3.1	1
C5H12O	1-pentanol	C	50.6	76.1	-25.5	0.0	3.1	1
C5H12O	2,2-dimethyl-1-propanol	C	31.1	50.0	-18.9	0.0	0.0	1
C5H12O	2-oxahexane	A	68.9	76.1	-7.2	0.0	3.1	1
C5H12O	3,3-dimethyl-2-oxabutane	A	46.2	50.0	-3.8	0.0	0.0	1
C5H12O	3-oxahexane	A	57.6	76.1	-18.5	0.0	3.1	1
C5H12O2	1,5-pentanediol	C	63.4	84.8	-21.4	0.0	4.2	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C5H12O4	2,2-bis(hydroxymethyl)-1,3-dihydroxypropane	B	87.2	84.8	2.4	0.0	4.2	2
C5H12O4	2,2-bis(hydroxymethyl)-1,3-dihydroxypropane	B	108.6	84.8	23.8	0.0	4.2	1
C5H12S	1-pentanethiol	A	88.8	76.1	12.6	0.0	3.1	1
C5H12S	2-methyl-2-butanethiol	A	53.8	58.7	-5.0	0.0	1.0	1
C5H12S	2-thiahexane	A	71.0	76.1	-5.1	0.0	3.1	1
C5H12S	3,3-dimethyl-2-thiabutane	A	44.1	50.0	-5.9	0.0	0.0	1
C5H12S	3-methyl-1-butanethiol	A	53.1	67.4	-14.3	0.0	2.1	1
C5H12S	3-methyl-2-butanethiol	A	53.0	58.7	-5.7	0.0	1.0	1
C5H12S	3-thiahexane	A	67.8	76.1	-8.3	0.0	3.1	1
C5H12S4	tetrakis(methylthia)methane	C	56.7	84.8	-28.1	0.0	4.2	1
C5H12Si	1,1-dimethyl-1-silacyclobutane	B	43.5	44.2	-0.8	0.7	0.0	2
C5H12Si	vinyltrimethylsilane	A	54.1	54.4	-0.3	0.0	0.5	2
C5H12Si	vinyltrimethylsilane	A	54.0	54.4	-0.3	0.0	0.5	2
C5H14N2	n,n-dimethyl-1,3-propanediamine	A	63.7	67.4	-3.7	0.0	2.1	1
C5H14N2O	tetramethylurea	A	49.2	63.1	-13.8	0.0	1.6	2
C6Cl3F3	1,3,5-trichloro-2,4,6-trifluorobenzene	B	59.4	35.1	24.3	1.8	0.0	1
C6Cl3F3	1,3,5-trichloro-2,4,6-trifluorobenzene	A	59.2	35.1	24.1	1.8	0.0	1
C6ClF5	pentafluorochlorobenzene	A	55.5	44.2	11.3	0.7	0.0	1
C6ClF5	pentafluorochlorobenzene	A	39.1	44.2	-5.1	0.7	0.0	1
C6D6	benzene-d6	B	35.0	29.3	5.6	2.5	0.0	1
C6F14	n-perfluorohexane	A	46.4	35.1	11.3	1.8	0.0	1
C6F15N	perfluorotriethylamine	A	29.8	76.1	-46.3	0.0	3.1	2
C6F15N	perfluorotriethylamine	A	46.3	76.1	-29.8	0.0	3.1	1
C6F5NO2	pentafluoronitrobenzene	A	47.1	50.0	-2.9	0.0	0.0	1
C6F6	hexafluorobenzene	A	41.7	29.3	12.3	2.5	0.0	1
C6F6	hexafluorobenzene	A	41.6	29.3	12.3	2.5	0.0	1
C6HF5	pentafluorobenzene	A	48.2	44.2	4.0	0.7	0.0	1
C6HF5	pentafluorobenzene	A	48.1	44.2	3.8	0.7	0.0	1
C6HF5O	pentafluorophenol	A	48.1	44.2	3.8	0.7	0.0	1
C6HF5O	pentafluorophenol	A	56.8	44.2	12.5	0.7	0.0	1
C6H2F4	1,2,3,4-tetrafluorobenzene	A	69.0	44.2	24.8	0.7	0.0	1
C6H2F4	1,2,3,5-tetrafluorobenzene	A	47.2	44.2	3.0	0.7	0.0	1
C6H2F4	1,2,4,5-tetrafluorobenzene	A	54.3	38.5	15.9	1.4	0.0	1
C6H2F5N	pentafluoroaniline	A	60.2	44.2	16.0	0.7	0.0	1
C6H3Cl4N	2-chloro-1-(trichloromethyl)pyridine	B	60.2	50.0	10.2	0.0	0.0	2
C6H3N3O6	1,3,5-trinitrobenzene	B	34.5	58.7	-24.2	0.0	1.0	1
C6H3N3O6	1,3,5-trinitrobenzene	B	37.7	58.7	-21.0	0.0	1.0	1
C6H3N3O6	1,3,5-trinitrobenzene	B	44.1	58.7	-14.6	0.0	1.0	1
C6H3N3O7	picric acid	B	43.4	58.7	-15.3	0.0	1.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C6H3N3O8	styphnic acid	B	73.6	50.0	23.6	0.0	0.0	2
C6H4Br2	1,2-dibromobenzene	D	49.4	44.2	5.2	0.7	0.0	2
C6H4Br2	1,3-dibromobenzene	D	53.4	44.2	9.2	0.7	0.0	2
C6H4Br2	1,4-dibromobenzene	D	57.0	38.5	18.5	1.4	0.0	2
C6H4Br2	1,4-dibromobenzene	C	55.7	38.5	17.2	1.4	0.0	1
C6H4Br2O	2,4-dibromophenol	D	47.9	50.0	-2.1	0.0	0.0	1
C6H4BrCl	2-chlorobromobenzene	D	47.5	50.0	-2.5	0.0	0.0	2
C6H4BrCl	3-chlorobromobenzene	D	48.8	50.0	-1.2	0.0	0.0	2
C6H4BrCl	4-chlorobromobenzene	D	55.5	44.2	11.3	0.7	0.0	2
C6H4BrI	2-bromoiodobenzene	D	52.5	50.0	2.5	0.0	0.0	2
C6H4BrI	3-bromoiodobenzene	D	46.2	50.0	-3.8	0.0	0.0	2
C6H4BrI	4-bromoiodobenzene	D	54.0	44.2	9.8	0.7	0.0	2
C6H4Cl2	1,2-dichlorobenzene	D	50.5	44.2	6.3	0.7	0.0	2
C6H4Cl2	1,3-dichlorobenzene	D	50.6	44.2	6.4	0.7	0.0	2
C6H4Cl2	1,4-dichlorobenzene	C	55.3	38.5	16.9	1.4	0.0	2
C6H4Cl2	1,4-dichlorobenzene	C	55.7	38.5	17.2	1.4	0.0	1
C6H4Cl2	1,4-dichlorobenzene	D	55.6	38.5	17.2	1.4	0.0	2
C6H4Cl2	1,4-dichlorobenzene	A	61.1	38.5	22.6	1.4	0.0	1
C6H4Cl2	1,4-dichlorobenzene	A	61.1	38.5	22.6	1.4	0.0	2
C6H4Cl2O	2,3-dichlorophenol	A	64.7	50.0	14.7	0.0	0.0	2
C6H4Cl2O	2,4-dichlorophenol	C	88.0	50.0	38.0	0.0	0.0	2
C6H4Cl2O	2,4-dichlorophenol	A	63.2	50.0	13.2	0.0	0.0	2
C6H4Cl2O	2,5-dichlorophenol	A	67.8	50.0	17.8	0.0	0.0	2
C6H4Cl2O	2,6-dichlorophenol	A	65.1	44.2	20.9	0.7	0.0	2
C6H4Cl2O	3,4-dichlorophenol	A	61.4	50.0	11.4	0.0	0.0	2
C6H4Cl2O	3,5-dichlorophenol	A	60.1	44.2	15.9	0.7	0.0	2
C6H4ClNO2	4-nitrochlorobenzene	C	33.4	50.0	-16.5	0.0	0.0	1
C6H4ClNO2	4-nitrochlorobenzene	C	50.4	50.0	0.4	0.0	0.0	2
C6H4F2	1,2-difluorobenzene	A	48.9	44.2	4.6	0.7	0.0	1
C6H4F2	1,3-difluorobenzene	A	46.5	44.2	2.3	0.7	0.0	1
C6H4I2	1,2-diiodobenzene	D	47.5	44.2	3.2	0.7	0.0	2
C6H4I2	1,3-diiodobenzene	D	51.9	44.2	7.6	0.7	0.0	2
C6H4I2	1,4-diiodobenzene	C	55.6	38.5	17.1	1.4	0.0	1
C6H4I2	1,4-diiodobenzene	D	55.6	38.5	17.2	1.4	0.0	2
C6H4N2O4	1,2-dinitrobenzene	C	58.3	54.4	4.0	0.0	0.5	1
C6H4N2O4	1,2-dinitrobenzene	C	57.7	54.4	3.3	0.0	0.5	2
C6H4N2O4	1,3-dinitrobenzene	C	47.8	54.4	-6.6	0.0	0.5	2
C6H4N2O4	1,3-dinitrobenzene	C	47.8	54.4	-6.6	0.0	0.5	1
C6H4N2O4	1,4-dinitrobenzene	C	62.9	54.4	8.6	0.0	0.5	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C6H4N2O4	1,4-dinitrobenzene	C	63.0	54.4	8.6	0.0	0.5	2
C6H4N2O5	2,3-dinitrophenol	A	62.9	54.4	8.6	0.0	0.5	2
C6H4N2O5	2,4-dinitrophenol	A	62.3	54.4	8.0	0.0	0.5	2
C6H4N2O5	2,5-dinitrophenol	A	62.3	54.4	7.9	0.0	0.5	2
C6H4N2O5	2,6-dinitrophenol	A	58.3	54.4	3.9	0.0	0.5	2
C6H4N2O5	3,4-dinitrophenol	A	62.4	54.4	8.0	0.0	0.5	2
C6H4O2	quinone	C	47.8	38.5	9.3	1.4	0.0	1
C6H5Br	bromobenzene	C	44.2	44.2	-0.1	0.7	0.0	1
C6H5Br	bromobenzene	A	43.8	44.2	-0.4	0.7	0.0	1
C6H5BrO	4-bromophenol	D	38.6	44.2	-5.7	0.7	0.0	1
C6H5BrO	4-bromophenol	C	49.3	44.2	5.1	0.7	0.0	2
C6H5Cl	chlorobenzene	C	41.9	44.2	-2.3	0.7	0.0	1
C6H5Cl3Si	phenyltrichlorosilane	A	50.0	40.9	9.1	1.1	0.0	1
C6H5ClO	2-chlorophenol	A	44.3	50.0	-5.8	0.0	0.0	2
C6H5ClO	3-chlorophenol	A	48.7	50.0	-1.3	0.0	0.0	2
C6H5ClO	4-chlorophenol	A	44.5	44.2	0.3	0.7	0.0	2
C6H5F	fluorobenzene	C	45.0	44.2	0.8	0.7	0.0	1
C6H5F	fluorobenzene	A	49.0	44.2	4.7	0.7	0.0	1
C6H5I	iodobenzene	C	40.3	44.2	-3.9	0.7	0.0	1
C6H5NO2	nitrobenzene	C	43.5	50.0	-6.5	0.0	0.0	1
C6H5NO2	nitrobenzene	C	38.8	50.0	-11.2	0.0	0.0	1
C6H5NO3	2-nitrophenol	A	54.9	50.0	4.9	0.0	0.0	2
C6H5NO3	3-nitrophenol	C	57.6	50.0	7.6	0.0	0.0	2
C6H5NO3	3-nitrophenol	A	51.9	50.0	1.9	0.0	0.0	2
C6H5NO3	4-nitrophenol	A	47.2	50.0	-2.8	0.0	0.0	2
C6H5NO3	4-nitrophenol	C	62.7	50.0	12.7	0.0	0.0	2
C6H5NO3	4-nitrophenol	C	52.3	50.0	2.3	0.0	0.0	2
C6H5NO3	4-nitrophenol	A	78.2	50.0	28.2	0.0	0.0	2
C6H6	benzene	A	35.4	29.3	6.1	2.5	0.0	1
C6H6	benzene	C	35.5	29.3	6.1	2.5	0.0	1
C6H6	benzene	C	33.3	29.3	4.0	2.5	0.0	2
C6H6	benzene	C	35.9	29.3	6.6	2.5	0.0	1
C6H6	benzene	C	35.2	29.3	5.9	2.5	0.0	1
C6H6	benzene	B	35.6	29.3	6.2	2.5	0.0	1
C6H6	benzene	C	32.1	29.3	2.8	2.5	0.0	1
C6H6N2O2	2-nitroaniline	C	47.0	50.0	-3.0	0.0	0.0	1
C6H6N2O2	3-nitroaniline	C	61.3	50.0	11.3	0.0	0.0	2
C6H6N2O2	3-nitroaniline	C	61.5	50.0	11.5	0.0	0.0	1
C6H6N2O2	4-nitroaniline	C	50.1	50.0	0.1	0.0	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C6H6N2O2	4-nitroaniline	C	50.3	50.0	0.3	0.0	0.0	1
C6H6O	phenol	A	36.7	44.2	-7.6	0.7	0.0	1
C6H6O	phenol	C	33.8	44.2	-10.4	0.7	0.0	2
C6H6O2	1,2-dihydroxybenzene	C	67.4	44.2	23.2	0.7	0.0	1
C6H6O2	1,3-dihydroxybenzene	B	53.8	44.2	9.6	0.7	0.0	2
C6H6O2	1,3-dihydroxybenzene	A	58.3	44.2	14.1	0.7	0.0	2
C6H6O2	1,3-dihydroxybenzene	C	55.6	44.2	11.4	0.7	0.0	1
C6H6O2	1,4-dihydroxybenzene	C	60.9	38.5	22.4	1.4	0.0	1
C6H6S	thiophenol	C	44.5	44.2	0.2	0.7	0.0	1
C6H6S	thiophenol	A	44.3	44.2	0.1	0.7	0.0	1
C6H7N	2-methylpyridine	A	47.1	50.0	-2.9	0.0	0.0	1
C6H7N	3-methylpyridine	A	55.6	50.0	5.6	0.0	0.0	1
C6H7N	4-methylpyridine	A	45.4	44.2	1.2	0.7	0.0	2
C6H7N	4-methylpyridine	A	45.5	44.2	1.2	0.7	0.0	2
C6H7N	aniline	C	39.6	44.2	-4.7	0.7	0.0	1
C6H7N	aniline	B	40.9	44.2	-3.4	0.7	0.0	1
C6H8	1,3-cyclohexadiene	B	26.1	44.2	-18.1	0.7	0.0	1
C6H8	1,4-cyclohexadiene	B	29.8	38.5	-8.7	1.4	0.0	1
C6H8N2	1,3-phenylenediamine	A	45.9	44.2	1.7	0.7	0.0	2
C6H8N2O2	1,3-dimethyluracil	B	58.9	50.0	8.9	0.0	0.0	2
C6H8O2	1,4-cyclohexanedione	A	32.2	50.0	-17.7	0.0	0.0	2
C6H8O4	dimethyl fumarate	C	93.7	76.1	17.6	0.0	3.1	1
C6H8O4	dimethyl maleate	C	57.6	76.1	-18.4	0.0	3.1	1
C6H8S	2,5-dimethylthiophene	A	38.9	44.2	-5.4	0.7	0.0	1
C6H9ClO2	chloroethyl methacrylate	A	72.3	76.1	-3.8	0.0	3.1	2
C6H9N	2,5-dimethylpyrrole	A	33.1	44.2	-11.1	0.7	0.0	2
C6H9N	2,5-dimethylpyrrole	A	33.1	44.2	-11.1	0.7	0.0	2
(C6H10O2)n	poly-epsilon-caprolactone	A	48.8	56.6	-7.8	0.0	6.8	2
C6H10	cyclohexene	C	48.8	44.2	4.6	0.7	0.0	1
C6H10	cyclohexene	A	49.9	44.2	5.6	0.7	0.0	1
C6H10	cyclohexene	A	50.1	44.2	5.8	0.7	0.0	1
C6H10N2O	2,3-diazabicyclo[2.2.2]oct-2-ene n-oxide	A	43.0	50.0	-7.0	0.0	0.0	2
C6H10O	cyclohexanone	A	44.6	50.0	-5.4	0.0	0.0	2
C6H10O	cyclohexene oxide	A	53.9	44.2	9.6	0.7	0.0	2
C6H10O2	epsilon-caprolactone	A	50.9	50.0	0.9	0.0	0.0	2
C6H10O2	epsilon-caprolactone	A	50.8	50.0	0.8	0.0	0.0	2
C6H11NO	epsilon-caprolactam	B	47.0	50.0	-3.0	0.0	0.0	2
C6H12	1-hexene	A	70.1	71.8	-1.7	0.0	2.6	1
C6H12	2,3-dimethyl-2-butene	A	50.4	38.5	11.9	1.4	0.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C6H12	2,3-dimethyl-2-butene	C	50.8	38.5	12.3	1.4	0.0	1
C6H12	3,3-dimethyl-1-butene	C	41.7	50.0	-8.3	0.0	0.0	1
C6H12	cyclohexane	A	45.3	35.1	10.2	1.8	0.0	1
C6H12	cyclohexane	C	42.2	35.1	7.1	1.8	0.0	1
C6H12	cyclohexane	B	46.4	35.1	11.3	1.8	0.0	1
C6H12	cyclohexane	A	45.8	35.1	10.7	1.8	0.0	1
C6H12	methylcyclopentane	A	53.0	50.0	3.0	0.0	0.0	1
C6H12	methylcyclopentane	C	52.9	50.0	2.9	0.0	0.0	1
C6H12N2	triethylenediamine	A	47.2	35.1	12.1	1.8	0.0	1
C6H12O	2-hexanone	A	68.4	76.1	-7.7	0.0	3.1	1
C6H12O	3,3-dimethyl-2-butanone	A	51.1	50.0	1.1	0.0	0.0	1
C6H12O	3-hexanone	A	66.7	71.8	-5.1	0.0	2.6	1
C6H12O	cyclohexanol	B	74.5	50.0	24.5	0.0	0.0	1
C6H12O	cyclohexanol	B	36.9	50.0	-13.1	0.0	0.0	1
C6H12O3	2,4,6-trimethyl-1,3,5-trioxane	C	54.4	35.1	19.3	1.8	0.0	1
C6H12O6	alpha-d-glucose	B	75.9	54.4	21.5	0.0	0.5	1
C6H12S	cyclohexanethiol	A	52.7	50.0	2.7	0.0	0.0	1
C6H12S	cyclopentyl-1-thiaethane	A	59.7	54.4	5.4	0.0	0.5	1
C6H13Br	1-bromohexane	C	96.0	84.8	11.2	0.0	4.2	1
C6H13N	2-methylpiperidine	A	69.0	50.0	19.0	0.0	0.0	2
C6H13N	2-methylpiperidine	A	69.0	50.0	19.0	0.0	0.0	2
C6H14	2,2-dimethylbutane	C	38.7	50.0	-11.2	0.0	0.0	1
C6H14	2,2-dimethylbutane	A	48.0	50.0	-2.0	0.0	0.0	2
C6H14	2,2-dimethylbutane	A	47.9	50.0	-2.1	0.0	0.0	1
C6H14	2,3-dimethylbutane	A	52.7	58.7	-6.0	0.0	1.0	1
C6H14	2,3-dimethylbutane	A	53.2	58.7	-5.5	0.0	1.0	1
C6H14	2-methylpentane	A	52.4	67.4	-15.0	0.0	2.1	1
* C6H14	2-methylpentane	C	3.1	67.4	-64.3	0.0	2.1	1
C6H14	3-methylpentane	A	48.1	67.4	-19.3	0.0	2.1	1
C6H14	n-hexane	C	73.3	76.1	-2.9	0.0	3.1	1
C6H14	n-hexane	A	73.6	76.1	-2.6	0.0	3.1	1
C6H14	n-hexane	C	70.4	76.1	-5.7	0.0	3.1	1
C6H14	n-hexane	C	69.4	76.1	-6.7	0.0	3.1	1
C6H14O	1-hexanol	B	68.1	84.8	-16.7	0.0	4.2	1
C6H14O	2,4-dimethyl-3-oxapentane	A	64.1	67.4	-3.3	0.0	2.1	1
C6H14O	2,4-dimethyl-3-oxapentane	C	59.2	67.4	-8.2	0.0	2.1	1
C6H14O	4-oxaheptane	A	63.9	84.8	-20.9	0.0	4.2	1
C6H14O2	1,6-hexanediol	A	79.6	93.5	-13.9	0.0	5.2	2
C6H14O3	2,5,8-trioxanonane	C	85.1	50.0	35.1	0.0	0.0	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C6H14O6	mannitol	B	123.7	93.5	30.1	0.0	5.2	1
C6H14S	1-hexanethiol	A	93.5	84.8	8.7	0.0	4.2	1
C6H14S	2,4-dimethyl-3-thiapentane	A	53.4	67.4	-14.0	0.0	2.1	1
C6H14S	4-thiaheptane	A	71.3	84.8	-13.5	0.0	4.2	1
C6H14S2	4,5-dithiaoctane	A	73.6	93.5	-19.9	0.0	5.2	1
C6H15Al	triethylaluminum	A	47.1	76.1	-29.0	0.0	3.1	2
C6H15As	triethylarsine	B	60.8	76.1	-15.2	0.0	3.1	1
C6H15B	triethylborane	B	65.7	76.1	-10.3	0.0	3.1	1
C6H15B	triethylboron	B	63.9	76.1	-12.1	0.0	3.1	2
C6H15Ga	triethylgallium	B	60.2	76.1	-15.9	0.0	3.1	1
C6H15In	triethylindium	B	54.8	76.1	-21.3	0.0	3.1	1
C6H15Sb	triethylstibine	B	61.4	76.1	-14.7	0.0	3.1	1
C6H16Si2	1,1,3,3-tetramethyl-1,3-disilacyclobutane	B	38.6	44.2	-5.7	0.7	0.0	2
C6H16Si2	tetramethyldisiletan	B	38.6	44.2	-5.7	0.7	0.0	1
C6H18BN	triethylamineborane	A	55.3	76.1	-20.8	0.0	3.1	1
C6H18O3Si3	hexamethylcyclotrisiloxane	B	49.6	35.1	14.5	1.8	0.0	1
C6H18OSi2	hexamethyldisiloxane	A	58.2	67.4	-9.2	0.0	2.1	1
C6H18Si2	hexamethyldisilane	C	54.4	35.1	19.3	1.8	0.0	2
C6H18Si2	hexamethyldisilane	B	54.5	35.1	19.4	1.8	0.0	1
C6H21N3Si3	hexamethyltrisilazane	A	59.6	35.1	24.5	1.8	0.0	2
C7F16	perfluoroheptane	A	68.3	102.2	-33.9	0.0	6.3	2
C7F8	octafluorotoluene	A	55.3	54.4	1.0	0.0	0.5	1
C7H3F5	2,3,4,5,6-pentafluorotoluene	A	54.5	44.2	10.3	0.7	0.0	1
C7H4MnNO3	azacymantrene	A	47.5	67.4	-19.9	0.0	2.1	2
C7H5Cl3	benzotrichloride	A	59.1	44.2	14.9	0.7	0.0	2
C7H5ClO2	2-chlorobenzoic acid	C	62.2	50.0	12.2	0.0	0.0	1
C7H5ClO2	3-chlorobenzoic acid	C	55.8	50.0	5.8	0.0	0.0	1
C7H5ClO2	4-chlorobenzoic acid	C	62.9	50.0	12.9	0.0	0.0	1
C7H5F3	benzotrifluoride	A	56.5	54.4	2.1	0.0	0.5	1
C7H5N	benzonitrile	A	42.2	44.2	-2.1	0.7	0.0	2
C7H5N	benzonitrile	A	42.2	44.2	-2.1	0.7	0.0	2
C7H5N	benzonitrile	A	42.2	44.2	-2.1	0.7	0.0	2
C7H5N	benzonitrile	A	42.2	44.2	-2.1	0.7	0.0	2
C7H5N3O6	2,4,5-trinitrotoluene	C	83.5	58.7	24.8	0.0	1.0	1
C7H5N5O8	nitramine	C	57.0	71.8	-14.8	0.0	2.6	1
C7H5NO4	2-nitrobenzoic acid	C	66.8	54.4	12.5	0.0	0.5	1
C7H5NO4	3-nitrobenzoic acid	C	46.7	54.4	-7.7	0.0	0.5	1
C7H5NS	benzothiazole	A	46.4	50.0	-3.6	0.0	0.0	1
C7H5NO4	4-nitrobenzoic acid	C	72.0	54.4	17.7	0.0	0.5	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C7H6F8O3	bis-(tetrafluoropropyl)carbonate	A	162.0	115.3	46.7	0.0	7.9	1
C7H6N2O4	2,4-dinitrotoluene	D	60.6	54.4	6.2	0.0	0.5	1
C7H6NO2	4-nitrobenzoic acid	C	72.0	54.4	17.7	0.0	0.5	2
C7H6O	benzaldehyde	A	43.2	50.0	-6.9	0.0	0.0	1
C7H6O2	benzoic acid	A	45.7	50.0	-4.3	0.0	0.0	1
C7H6O2	benzoic acid	C	43.9	50.0	-6.2	0.0	0.0	1
C7H6O2	benzoic acid	A	45.5	50.0	-4.5	0.0	0.0	1
C7H6O2	benzoic acid	A	45.5	50.0	-4.5	0.0	0.0	1
C7H6O2	benzoic acid	D	44.1	50.0	-6.0	0.0	0.0	1
C7H6O2	benzoic acid	C	41.1	50.0	-8.9	0.0	0.0	1
C7H7F	4-fluorotoluene	A	43.2	44.2	-1.1	0.7	0.0	1
C7H7NO2	2-aminobenzoic acid	C	48.8	50.0	-1.2	0.0	0.0	1
C7H7NO2	3-aminobenzoic acid	C	48.2	50.0	-1.8	0.0	0.0	1
C7H7NO2	4-aminobenzoic acid	C	45.3	50.0	-4.7	0.0	0.0	1
* C7H7NO2	4-nitrotoluene	B	222.2	50.0	172.2	0.0	0.0	1
C7H7NO2	4-nitrotoluene	A	51.8	50.0	1.8	0.0	0.0	1
C7H8	cycloheptatriene	A	21.1	28.1	-7.0	2.6	0.0	1
C7H8	toluene	B	36.8	44.2	-7.5	0.7	0.0	1
C7H8	toluene	A	37.2	44.2	-7.0	0.7	0.0	1
C7H8	toluene	A	37.2	44.2	-7.1	0.7	0.0	1
C7H8N2O	phenylurea	A	56.3	58.7	-2.4	0.0	1.0	2
C7H8O	2-methylphenol	A	46.0	50.0	-4.0	0.0	0.0	2
C7H8O	3-methylphenol	A	33.0	50.0	-16.9	0.0	0.0	2
C7H8O	4-methylphenol	C	39.6	44.2	-4.6	0.7	0.0	2
C7H8O	4-methylphenol	A	38.5	44.2	-5.8	0.7	0.0	2
C7H8O	anisole	C	58.1	54.4	3.7	0.0	0.5	2
C7H8O	anisole	A	48.0	54.4	-6.4	0.0	0.5	2
C7H8O	benzyl alcohol	C	34.8	54.4	-19.5	0.0	0.5	1
C7H8O	m-hydroxytoluene	A	37.5	50.0	-12.4	0.0	0.0	1
C7H8O	o-hydroxytoluene	A	52.0	50.0	2.0	0.0	0.0	1
C7H8O	p-hydroxytoluene	A	41.3	44.2	-3.0	0.7	0.0	1
C7H8S	methyl phenyl sulfide	A	57.9	50.0	7.9	0.0	0.0	1
C7H9N	4-methylaniline	C	54.8	44.2	10.5	0.7	0.0	2
C7H10N2O	6,7-diazatricyclo[3.2.2.0(2,4)]non-6-ene n-oxide	A	48.7	50.0	-1.3	0.0	0.0	2
C7H10N2O2	1,3,6-trimethyluracil	B	55.1	50.0	5.1	0.0	0.0	2
C7H12	cis-cycloheptene	A	53.1	50.0	3.1	0.0	0.0	2
C7H12O2	butyl acrylate	A	82.6	84.8	-2.2	0.0	4.2	2
C7H13N	1-azabicyclo[2.2.2]octane	A	40.3	40.9	-0.6	1.1	0.0	1
C7H13NO	zeta-enantholactam	B	44.4	50.0	-5.6	0.0	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C7H14	1,1-dimethylcyclopentane	A	49.5	44.2	5.3	0.7	0.0	1
C7H14	1,2-dimethylcyclopentane	C	41.6	44.2	-2.6	0.7	0.0	1
C7H14	1-cis-2-dimethylcyclopentane	A	54.7	50.0	4.7	0.0	0.0	1
C7H14	1-heptene	A	80.4	80.5	-0.1	0.0	3.7	1
C7H14	1-heptene	C	82.5	80.5	2.1	0.0	3.7	1
C7H14	1-trans-3-dimethylcyclopentane	A	53.0	44.2	8.8	0.7	0.0	1
C7H14	cycloheptane	A	47.5	50.0	-2.5	0.0	0.0	1
C7H14	ethylcyclopentane	A	51.0	54.4	-3.4	0.0	0.5	1
C7H14	methylcyclohexane	A	46.1	50.0	-4.0	0.0	0.0	1
C7H14	methylcyclohexane	C	45.6	50.0	-4.4	0.0	0.0	1
C7H14O	2,4-dimethyl-3-pentanone	A	54.6	63.1	-8.5	0.0	1.6	1
C7H14O	cycloheptanol	A	29.2	50.0	-20.8	0.0	0.0	1
C7H14O	heptanal	C	102.6	89.2	13.5	0.0	4.7	1
C7H14O2	heptanoic acid	B	67.1	89.2	-22.0	0.0	4.7	1
C7H16	2,2,3-trimethylbutane	C	28.5	50.0	-21.4	0.0	0.0	1
C7H16	2,2-dimethylpentane	C	39.6	58.7	-19.1	0.0	1.0	1
C7H16	2,2-dimethylpentane	A	39.0	58.7	-19.7	0.0	1.0	1
C7H16	2,4-dimethylpentane	C	43.9	67.4	-23.5	0.0	2.1	1
C7H16	2,4-dimethylpentane	A	44.5	67.4	-22.9	0.0	2.1	1
C7H16	2-methylhexane	A	59.3	76.1	-16.8	0.0	3.1	1
C7H16	2-methylhexane	C	57.6	76.1	-18.5	0.0	3.1	1
C7H16	2-methylhexane	C	57.6	76.1	-18.5	0.0	3.1	1
C7H16	3,3-dimethylpentane	C	51.1	67.4	-16.2	0.0	2.1	1
C7H16	3,3-dimethylpentane	A	55.3	67.4	-12.1	0.0	2.1	2
C7H16	3-ethylpentane	A	61.8	76.1	-14.3	0.0	3.1	1
C7H16	3-ethylpentane	C	61.3	76.1	-14.8	0.0	3.1	1
C7H16	n-heptane	B	76.8	84.8	-8.0	0.0	4.2	1
C7H16	n-heptane	B	76.8	84.8	-8.0	0.0	4.2	1
C7H16	n-heptane	A	76.9	84.8	-8.0	0.0	4.2	1
C7H16	n-heptane	A	77.0	84.8	-7.9	0.0	4.2	1
C7H16	n-heptane	A	76.9	84.8	-8.0	0.0	4.2	1
C7H16	n-heptane	A	76.8	84.8	-8.0	0.0	4.2	2
C7H16	n-heptane	B	77.0	84.8	-7.9	0.0	4.2	1
C7H16	n-heptane	A	76.9	84.8	-7.9	0.0	4.2	1
C7H16	n-heptane	C	77.7	84.8	-7.1	0.0	4.2	1
C7H16	n-heptane	A	76.8	84.8	-8.0	0.0	4.2	1
C7H16	n-heptane	C	77.7	84.8	-7.1	0.0	4.2	1
C7H16O	1-heptanol	C	75.6	93.5	-17.9	0.0	5.2	1
C7H16S	1-heptanethiol	A	110.4	93.5	16.9	0.0	5.2	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C7H17NSi	n-(beta-trimethylsilylethyl)ethylenimine	A	60.2	54.4	5.8	0.0	0.5	2
C7H20Si2	hexamethyldisilylmethane	B	79.0	58.7	20.3	0.0	1.0	2
C7H20Si2	hexamethyldisilylmethane	A	79.0	58.7	20.3	0.0	1.0	2
C8D8	styrene-d8	B	44.3	50.0	-5.7	0.0	0.0	2
C8H4N2	1,2-dicyanobenzene	A	48.3	44.2	4.1	0.7	0.0	2
C8H4N2	1,2-dicyanobenzene	A	48.3	44.2	4.1	0.7	0.0	2
C8H5MnO3	cymantrene	A	55.1	63.1	-7.9	0.0	1.6	2
C8H6	phenylacetylene	A	41.5	44.2	-2.8	0.7	0.0	2
C8H6S	benzothiophene	A	38.8	50.0	-11.1	0.0	0.0	1
C8H7ClN2O2	2-chloroisonitrosoacetanilide	D	69.2	67.4	1.8	0.0	2.1	2
C8H7N5O8	2,4,6-trinitro-n-(methylnitro)-m-toluidine	C	51.5	76.1	-24.6	0.0	3.1	1
C8H7N5O8	2,4,6-trinitrophenylethyl nitramine	C	63.7	80.5	-16.7	0.0	3.7	1
C8H8	cyclooctatetraene	A	42.0	27.0	15.0	2.8	0.0	1
C8H8	styrene	B	45.2	50.0	-4.9	0.0	0.0	2
C8H8	styrene	A	45.2	50.0	-4.8	0.0	0.0	2
C8H8	styrene	A	45.2	50.0	-4.9	0.0	0.0	1
C8H8N2O2	isonitrosoacetanilide	D	23.2	67.4	-44.2	0.0	2.1	2
C8H8O2	m-toluic acid	C	41.2	50.0	-8.8	0.0	0.0	1
C8H8O2	o-toluic acid	C	53.5	50.0	3.5	0.0	0.0	1
C8H8O2	p-toluic acid	C	50.2	50.0	0.2	0.0	0.0	1
C8H9NO	acetanilide	A	55.9	58.7	-2.8	0.0	1.0	1
C8H9NO2	methyl phenylcarbamate	B	44.8	58.7	-13.9	0.0	1.0	1
C8H9NO2	o-hydroxyacetanilide	C	58.3	58.7	-0.4	0.0	1.0	1
C8H10	1,2-dimethylbenzene	A	54.9	44.2	10.6	0.7	0.0	1
C8H10	1,2-dimethylbenzene	C	52.8	44.2	8.5	0.7	0.0	1
C8H10	1,3-dimethylbenzene	C	53.4	44.2	9.1	0.7	0.0	1
C8H10	1,3-dimethylbenzene	A	51.4	44.2	7.1	0.7	0.0	1
C8H10	1,4-dimethylbenzene	A	59.8	38.5	21.3	1.4	0.0	2
C8H10	1,4-dimethylbenzene	A	59.7	38.5	21.3	1.4	0.0	1
C8H10	1,4-dimethylbenzene	A	59.8	38.5	21.3	1.4	0.0	1
C8H10	1,4-dimethylbenzene	C	59.1	38.5	20.7	1.4	0.0	1
C8H10	ethylbenzene	A	51.4	54.4	-2.9	0.0	0.5	1
C8H10	ethylbenzene	A	51.5	54.4	-2.8	0.0	0.5	1
C8H10	ethylbenzene	C	51.5	54.4	-2.9	0.0	0.5	1
C8H10N4O2	caffeine	B	13.1	50.0	-36.8	0.0	0.0	2
C8H10N4O2	caffeine, anhydrous	B	52.2	50.0	2.2	0.0	0.0	1
C8H10O	2,3-dimethylphenol	A	60.8	50.0	10.8	0.0	0.0	2
C8H10O	2,5-dimethylphenol	A	67.2	50.0	17.2	0.0	0.0	2
C8H10O	2,6-dimethylphenol	A	59.3	44.2	15.0	0.7	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C8H10O	3,4-dimethylphenol	A	54.3	50.0	4.3	0.0	0.0	2
C8H10O	3,5-dimethylphenol	A	53.4	44.2	9.2	0.7	0.0	2
C8H11N	2-cyanobicyclo[2.2.1]heptane(endo)	A	21.6	50.0	-28.3	0.0	0.0	1
C8H11N	2-cyanobicyclo[2.2.1]heptane(exo)	A	43.2	50.0	-6.8	0.0	0.0	1
* (C8H12)n	polyoctadiene	B	1.1	52.2	-51.1	0.0	6.3	1
C8H12	bicyclo[2.2.2]octene-2	B	47.5	44.2	3.3	0.7	0.0	1
C8H12	cycloocta-1,5-diene	A	46.2	44.2	1.9	0.7	0.0	2
C8H12	cycloocta-1,5-diene	A	46.2	44.2	2.0	0.7	0.0	1
C8H12N2O2	1,6-hexamethylene diisocyanate	A	90.4	93.5	-3.1	0.0	5.2	2
C8H12N2O2	1,6-hexamethylene diisocyanate	A	90.4	93.5	-3.1	0.0	5.2	2
(C8H14)n	polyoctenylene	A	89.6	61.0	28.7	0.0	7.3	2
C8H14	2-methylbicyclo[2.2.1]heptane(endo)	A	36.7	50.0	-13.2	0.0	0.0	1
C8H14	2-methylbicyclo[2.2.1]heptane(exo)	A	51.1	50.0	1.1	0.0	0.0	1
C8H14	bicyclo[2.2.2]octane	B	46.6	35.1	11.5	1.8	0.0	1
C8H14	cyclooctene	A	10.3	50.0	-39.6	0.0	0.0	2
C8H14O	3-oxabicyclo[3.2.2]nonane	A	48.7	50.0	-1.3	0.0	0.0	1
C8H14O2	butyl methacrylate	A	70.9	84.8	-13.9	0.0	4.2	2
C8H15N	3-azabicyclo[3.2.2]nonane	A	14.8	50.0	-35.1	0.0	0.0	1
C8H15NO2	dimethylaminoethyl methacrylate	A	70.9	84.8	-13.9	0.0	4.2	2
C8H16	1,1-dimethylcyclohexane	A	47.5	50.0	-2.5	0.0	0.0	1
C8H16	1-cis-2-dimethylcyclohexane	A	55.2	50.0	5.2	0.0	0.0	1
C8H16	1-cis-3-dimethylcyclohexane	A	54.8	50.0	4.8	0.0	0.0	1
C8H16	1-cis-4-dimethylcyclohexane	A	50.1	50.0	0.1	0.0	0.0	1
C8H16	1-octene	A	89.3	89.2	0.1	0.0	4.7	1
C8H16	1-trans-2-dimethylcyclohexane	A	56.7	50.0	6.7	0.0	0.0	1
C8H16	1-trans-3-dimethylcyclohexane	A	53.9	50.0	3.9	0.0	0.0	1
C8H16	1-trans-4-dimethylcyclohexane	A	52.2	44.2	8.0	0.7	0.0	1
C8H16	2,4,4-trimethyl-1-pentene	C	49.0	54.4	-5.4	0.0	0.5	1
C8H16	2,4,4-trimethyl-2-pentene	C	40.9	50.0	-9.1	0.0	0.0	1
C8H16	cyclooctane	A	48.8	50.0	-1.2	0.0	0.0	1
C8H16	ethylcyclohexane	A	51.5	54.4	-2.9	0.0	0.5	1
C8H16	ethylcyclohexane	C	51.3	54.4	-3.1	0.0	0.5	1
C8H16	n-propylcyclopentane	A	64.4	63.1	1.3	0.0	1.6	1
C8H16O	2-octanone	A	96.6	89.2	7.4	0.0	4.7	1
C8H16O	octanal	B	104.4	97.9	6.5	0.0	5.8	2
C8H16O2	butyl butanoate	A	82.2	97.9	-15.7	0.0	5.8	2
C8H16O2	hexyl ethanoate	A	93.5	97.9	-4.4	0.0	5.8	2
C8H16O2	octanoic acid	B	73.7	97.9	-24.1	0.0	5.8	1
C8H16O2	octanoic acid	B	73.9	97.9	-24.0	0.0	5.8	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C8H18	2,2,3,3-tetramethylbutane	A	33.3	44.2	-10.9	0.7	0.0	1
C8H18	2,2,4-trimethylpentane	A	55.6	58.7	-3.2	0.0	1.0	1
C8H18	2,2,4-trimethylpentane	C	54.7	58.7	-4.0	0.0	1.0	1
C8H18	2,3,4-trimethylpentane	B	56.7	67.4	-10.7	0.0	2.1	1
C8H18	2-methylheptane	A	72.6	84.8	-12.2	0.0	4.2	1
C8H18	3-methylheptane	A	76.6	84.8	-8.3	0.0	4.2	1
C8H18	n-octane	C	95.7	93.5	2.2	0.0	5.2	1
C8H18	n-octane	C	93.2	93.5	-0.4	0.0	5.2	1
C8H18	n-octane	A	95.8	93.5	2.3	0.0	5.2	1
C8H18N2	1,1-dimethylazoethane	A	59.9	76.1	-16.2	0.0	3.1	2
C8H18N2O	1,1-dimethylazoxyethane	A	71.1	76.1	-5.1	0.0	3.1	2
C8H18N2O2	bis-hydroxyethylpiperazine	B	64.0	80.5	-16.5	0.0	3.7	2
C8H18O4	2,5,8,11-tetraoxadodecane	C	103.4	128.4	-24.9	0.0	9.4	1
C8H18S	5-thianonane	A	98.1	102.2	-4.2	0.0	6.3	1
C8H19NSi	n-(beta-trimethylsilylethyl)trimethylenimine	A	64.7	63.1	1.6	0.0	1.6	2
C8H20Ge	tetraethylgermane	A	68.4	84.8	-16.4	0.0	4.2	2
C8H20Ge	tetraethylgermane	B	68.7	84.8	-16.1	0.0	4.2	2
C8H20Ge	tetraethylgermane	B	70.0	84.8	-14.8	0.0	4.2	1
C8H20Pb	tetraethyl lead	A	64.4	84.8	-20.4	0.0	4.2	2
C8H20Pb	tetraethyl lead	B	61.5	84.8	-23.3	0.0	4.2	2
C8H20Si	tetraethyl silicon	B	68.7	84.8	-16.1	0.0	4.2	2
C8H20Si	tetraethylsilane	B	70.4	84.8	-14.4	0.0	4.2	1
C8H20Sn	tetraethyl tin	B	64.4	84.8	-20.4	0.0	4.2	2
C8H20Sn	tetraethylstannane	B	54.0	84.8	-30.8	0.0	4.2	1
C8H24Si4O4	octamethyltetrasiloxane	A	81.9	50.0	31.9	0.0	0.0	2
C8H28Ni4Si	octamethyltetrasilazane	A	68.1	50.0	18.1	0.0	0.0	2
C9H7N	isoquinoline	A	45.2	50.0	-4.8	0.0	0.0	2
C9H7N	quinoline	A	41.6	50.0	-8.4	0.0	0.0	2
C9H7N	quinoline	C	41.8	50.0	-8.2	0.0	0.0	1
C9H7O5	trimellitic anhydride	D	27.2	50.0	-22.8	0.0	0.0	1
C9H8	indene	A	37.6	50.0	-12.4	0.0	0.0	1
C9H8O2	cinnamic acid	A	55.7	58.7	-3.0	0.0	1.0	2
C9H10	alpha-methylstyrene	B	47.5	50.0	-2.5	0.0	0.0	2
C9H10	indan	A	38.8	44.2	-5.5	0.7	0.0	1
C9H10N2O3	2-methoxyisonitrosoacetanilide	D	65.9	76.1	-10.2	0.0	3.1	2
C9H10N2O3	4-methoxyisonitrosoacetanilide	D	18.1	76.1	-58.0	0.0	3.1	2
C9H10O2	phenylpropionic acid	C	48.4	67.4	-19.0	0.0	2.1	2
C9H11N	1,2,3,4-tetrahydroquinoline	A	40.8	50.0	-9.3	0.0	0.0	2
C9H11N	5,6,7,8-tetrahydroquinoline	A	40.8	50.0	-9.3	0.0	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C9H11NO2	ethyl phenylcarbamate	B	49.9	67.4	-17.5	0.0	2.1	1
C9H12	1,2,3-trimethylbenzene	A	41.8	44.2	-2.4	0.7	0.0	1
C9H12	1,2,4-trimethylbenzene	A	57.5	50.0	7.5	0.0	0.0	1
C9H12	1,2,4-trimethylbenzene	C	55.3	50.0	5.3	0.0	0.0	1
C9H12	1,3,5-trimethylbenzene	A	41.7	35.1	6.6	1.8	0.0	1
C9H12	isopropylbenzene	A	41.4	54.4	-12.9	0.0	0.5	1
C9H12	n-propylbenzene	A	53.4	63.1	-9.7	0.0	1.6	1
C9H14O2	carbopropoxy methyl methacrylate	A	71.9	97.9	-26.0	0.0	5.8	2
C9H14O6	triacetin	A	93.7	106.6	-12.8	0.0	6.8	2
C9H16	cis-hexahydroindan	A	53.4	50.0	3.4	0.0	0.0	1
C9H16	trans-hexahydroindan	A	51.0	50.0	1.0	0.0	0.0	1
C9H18	n-butylcyclopentane	A	68.5	71.8	-3.3	0.0	2.6	1
C9H18	n-propylcyclohexane	A	73.3	63.1	10.3	0.0	1.6	1
C9H18O	5-nonanone	A	96.0	97.9	-1.9	0.0	5.8	1
C9H18O2	amyl butyrate	A	99.8	106.6	-6.8	0.0	6.8	2
C9H18O2	butyl pentanoic acid	A	92.9	106.6	-13.6	0.0	6.8	2
C9H18O2	nonanoic acid	B	100.4	106.6	-6.2	0.0	6.8	1
C9H18O2	nonanoic acid	B	91.8	106.6	-14.8	0.0	6.8	1
C9H20	2,2,3,3-tetramethylpentane	A	50.8	58.7	-7.9	0.0	1.0	2
C9H20	2,2,4,4-tetramethylpentane	A	47.2	50.0	-2.8	0.0	0.0	2
C9H20	3,3-diethylpentane	B	47.8	84.8	-36.9	0.0	4.2	2
C9H20	3,3-diethylpentane	A	48.2	84.8	-36.6	0.0	4.2	2
C9H20	n-nonane	C	100.9	102.2	-1.3	0.0	6.3	1
C9H20	n-nonane	A	99.3	102.2	-2.9	0.0	6.3	1
C9H20N2O	n,n'-dibutylurea	A	78.5	115.3	-36.8	0.0	7.9	2
C9H24Si2	1,3-bis(trimethylsilyl)propane	B	71.8	67.4	4.4	0.0	2.1	2
C9H24Si3	1,1,3,3,5,5-hexamethyl-1,3,5-trisilacyclohexane	B	61.3	35.1	26.2	1.8	0.0	2
C10F16	perfluorobicyclo[4.4.0]dec-1,6-diene	A	48.4	50.0	-1.6	0.0	0.0	1
C10F18	cis-perfluorodecalin	B	56.9	44.2	12.7	0.7	0.0	1
C10F18	trans-perfluorodecalin	B	61.0	44.2	16.7	0.7	0.0	1
C10H2O6	pyromellitic dianhydride	D	28.4	38.5	-10.0	1.4	0.0	1
C10H6OS2	naphthalene-1,8-disulfide-s-oxide	B	36.9	44.2	-7.4	0.7	0.0	1
C10H7Br	2-bromonaphthalene	A	61.9	50.0	11.9	0.0	0.0	1
C10H8	naphthalene	C	53.9	38.5	15.4	1.4	0.0	1
C10H8	naphthalene	B	54.4	38.5	15.9	1.4	0.0	1
C10H8	naphthalene	C	53.2	38.5	14.7	1.4	0.0	1
C10H8	naphthalene	A	53.9	38.5	15.4	1.4	0.0	1
C10H8	naphthalene	C	53.2	38.5	14.8	1.4	0.0	1
C10H8	naphthalene	C	53.8	38.5	15.3	1.4	0.0	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C10H8	naphthalene	A	53.7	38.5	15.2	1.4	0.0	1
C10H8	naphthalene	C	54.0	38.5	15.6	1.4	0.0	1
C10H8	naphthalene	D	54.5	38.5	16.1	1.4	0.0	1
C10H8	naphthalene	C	53.7	38.5	15.2	1.4	0.0	1
C10H8	naphthalene	A	51.6	38.5	13.1	1.4	0.0	1
C10H8O	alpha-naphthol	C	62.9	50.0	12.9	0.0	0.0	1
C10H8O	alpha-naphthol	C	63.7	50.0	13.7	0.0	0.0	1
C10H8O	beta-naphthol	C	47.7	50.0	-2.3	0.0	0.0	1
C10H10	bullvalene	A	41.6	44.2	-2.6	0.7	0.0	2
C10H10N2O2	n-ethanol isatoxine	D	57.2	67.4	-10.1	0.0	2.1	2
C10H10O4	dimethyl o-phthalate	B	61.8	71.8	-10.0	0.0	2.6	2
C10H10O4	dimethyl o-phthalate	C	61.8	71.8	-10.0	0.0	2.6	1
C10H10O4	dimethyl terephthalate	B	76.4	71.8	4.7	0.0	2.6	1
C10H10O4	dimethyl terephthalate	B	77.6	71.8	5.8	0.0	2.6	1
C10H12	1,2,3,4-tetrahydronaphthalene	A	52.4	44.2	8.2	0.7	0.0	1
C10H12	endo-dicyclopentadiene	A	52.0	50.0	2.0	0.0	0.0	2
C10H12N2O3	2-ethoxyisonitrosoacetanilide	D	56.8	84.8	-28.0	0.0	4.2	2
C10H12N2O3	4-ethoxyisonitrosoacetanilide	D	15.5	84.8	-69.3	0.0	4.2	2
C10H12N2S	n-allyl-n'-phenylthiourea	B	73.6	80.5	-6.8	0.0	3.7	2
C10H13NO2	propyl phenylcarbamate	B	63.7	76.1	-12.4	0.0	3.1	1
C10H14	1,2,3,4-tetramethylbenzene	C	42.3	44.2	-1.9	0.7	0.0	1
C10H14	1,2,3,5-tetramethylbenzene	C	52.0	44.2	7.8	0.7	0.0	1
C10H14	1,2,4,5-tetramethylbenzene	C	59.3	38.5	20.8	1.4	0.0	1
C10H14	n-butylbenzene	C	59.5	71.8	-12.3	0.0	2.6	1
C10H14	n-butylbenzene	A	60.8	71.8	-10.9	0.0	2.6	1
C10H14	p-cymene	C	47.3	54.4	-7.0	0.0	0.5	1
C10H14	tert-butylbenzene	C	39.1	40.9	-1.8	1.1	0.0	1
C10H14Si	vinyldimethylphenylsilane	A	64.3	58.7	5.6	0.0	1.0	2
C10H14Si	vinyldimethylphenylsilane	A	64.5	58.7	5.8	0.0	1.0	2
C10H15NO	carvoxime(dl)	B	46.6	54.4	-7.7	0.0	0.5	1
C10H15NO	carvoxime(l)	B	65.5	54.4	11.2	0.0	0.5	1
(C10H16O4)n	polybutylene glycol adipate	B	75.4	95.8	-20.3	0.0	11.5	2
C10H16	tricyclo[5.2.1.0^(2,6)]decane	C	8.4	40.9	-32.4	1.1	0.0	1
C10H16O	camphor	C	15.1	40.9	-25.7	1.1	0.0	1
C10H18	cis-decahydronaphthalene	B	48.6	50.0	-1.4	0.0	0.0	2
C10H18	cis-decahydronaphthalene	A	59.4	50.0	9.4	0.0	0.0	1
C10H18	trans-decahydronaphthalene	B	52.3	50.0	2.3	0.0	0.0	2
C10H18	trans-decahydronaphthalene	A	51.1	50.0	1.1	0.0	0.0	1
C10H19NO2	diethylaminoethyl methacrylate	A	63.0	102.2	-39.2	0.0	6.3	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C10H20	1-decene	A	106.8	106.6	0.2	0.0	6.8	1
C10H20	2,2,5,5-tetramethylhex-3-ene	A	61.1	58.7	2.3	0.0	1.0	2
C10H20	n-butylcyclohexane	A	71.4	71.8	-0.4	0.0	2.6	1
C10H20O	decanal	B	113.5	115.3	-1.8	0.0	7.9	2
C10H20O2	decanoic acid	B	92.0	115.3	-23.3	0.0	7.9	1
C10H20O2	decanoic acid	B	91.3	115.3	-24.0	0.0	7.9	1
C10H20O2	decanoic acid	C	97.4	115.3	-17.9	0.0	7.9	2
C10H22	2-methylnonane	C	88.0	102.2	-14.2	0.0	6.3	1
C10H22	3-methylnonane(dl)	C	99.2	102.2	-3.1	0.0	6.3	1
C10H22	4-methylnonane(dl)	C	86.9	102.2	-15.3	0.0	6.3	1
C10H22	5-methylnonane	C	89.1	102.2	-13.1	0.0	6.3	1
C10H22	n-decane	C	118.4	111.0	7.4	0.0	7.3	1
C10H22	n-decane	A	117.9	111.0	7.0	0.0	7.3	1
C10H22S	1-decanethiol	A	134.4	119.7	14.7	0.0	8.4	1
C10H26O3Si3	1,1,3,3-tetraethyl-5,5-dimethylcyclotrisiloxane	A	37.3	80.5	-43.2	0.0	3.7	2
C11H10	1-methylnaphthalene	A	49.3	50.0	-0.7	0.0	0.0	1
C11H10	2-methylnaphthalene	C	39.0	50.0	-11.0	0.0	0.0	1
C11H10	2-methylnaphthalene	A	59.3	50.0	9.3	0.0	0.0	1
C11H10O4	p-methacryloyloxybenzoic acid	A	74.7	58.7	16.0	0.0	1.0	2
C11H16	pentamethylbenzene	C	32.5	44.2	-11.7	0.7	0.0	1
C11H16Si	vinyldimethylbenzylsilane	A	56.8	67.4	-10.5	0.0	2.1	2
C11H20O2	undecanolactone	A	59.2	50.0	9.2	0.0	0.0	1
C11H20O2	undecanolactone	A	59.2	50.0	9.2	0.0	0.0	1
C11H20O3Si3	1,1,1,3,5,5,5-heptamethyl-3-phenyltrisiloxane	A	80.7	50.0	30.7	0.0	0.0	2
C11H22	1-undecene	A	118.3	115.3	2.9	0.0	7.9	1
C11H22O2	undecanoic acid	B	109.8	124.0	-14.1	0.0	8.9	1
C11H22O2	undecanoic acid	B	114.2	124.0	-9.8	0.0	8.9	1
C11H24	2-methyldecane	A	111.9	111.0	0.9	0.0	7.3	1
C11H24	n-undecane	A	118.6	119.7	-1.1	0.0	8.4	1
C11H24	n-undecane	C	117.1	119.7	-2.6	0.0	8.4	1
C11H24O	2-oxadodecane	A	130.3	128.4	1.9	0.0	9.4	1
* C12H7Cl2NO3	2,4-dichloro-4'-nitrodiphenyl ether	C	7.9	63.1	-55.1	0.0	1.6	2
C12H8	acenaphthylene	C	19.1	44.2	-25.1	0.7	0.0	1
C12H8Cl2O2S	4,4'-dichlorodiphenyl sulphone	A	57.8	54.4	3.5	0.0	0.5	2
C12H8Cl6Si2	o,o'-bis-trichlorosilylbiphenyl	A	61.3	50.0	11.3	0.0	0.0	2
C12H8N2O5	4,4'-dinitrodiphenyl ether	D	24.6	67.4	-42.7	0.0	2.1	1
C12H8OS2	diphenylene-2,2'-disulfide-s-oxide	B	44.2	44.2	0.0	0.7	0.0	1
C12H8S	dibenzothiophene	A	58.2	44.2	13.9	0.7	0.0	2
C12H9Cl	2-chlorobiphenyl	A	47.6	50.0	-2.4	0.0	0.0	1

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C12H9Cl	2-chlorobiphenyl	B	47.6	50.0	-2.4	0.0	0.0	1
	C12H9Cl	4-chlorobiphenyl	B	38.2	50.0	-11.8	0.0	0.0	1
	C12H9Cl	4-chlorobiphenyl	A	38.2	50.0	-11.7	0.0	0.0	2
	C12H9Cl3Si	o-trichlorosilylbiphenyl	B	61.1	58.7	2.4	0.0	1.0	1
	C12H9Cl3Si	p-trichlorosilylbiphenyl	B	49.8	58.7	-8.9	0.0	1.0	1
	C12H9Cl3Si	p-trichlorosilylbiphenyl	A	49.8	58.7	-8.9	0.0	1.0	2
	C12H9N	carbazole	B	52.2	44.2	8.0	0.7	0.0	1
	C12H10	acenaphthene	C	55.2	44.2	11.0	0.7	0.0	1
	C12H10	acenaphthene	A	58.5	44.2	14.3	0.7	0.0	1
*	C12H10	acenaphthene	C	68.2	44.2	24.0	0.7	0.0	1
	C12H10	biphenyl	B	54.6	50.0	4.6	0.0	0.0	1
*	C12H10	biphenyl	A	6.5	50.0	-43.4	0.0	0.0	2
	C12H10	biphenyl	A	54.3	50.0	4.3	0.0	0.0	2
	C12H10	biphenyl	B	58.0	50.0	8.0	0.0	0.0	2
	C12H10	biphenyl	C	54.4	50.0	4.4	0.0	0.0	1
	C12H10	biphenyl	C	60.3	50.0	10.3	0.0	0.0	2
	C12H10	biphenyl	C	54.2	50.0	4.2	0.0	0.0	1
	C12H10	biphenyl	A	54.3	50.0	4.3	0.0	0.0	2
	C12H10N2	azobenzene	C	65.4	44.2	21.2	0.7	0.0	2
	C12H10N2	trans-azobenzene	A	66.1	44.2	21.8	0.7	0.0	2
	C12H10N2	trans-azobenzene	B	66.0	44.2	21.8	0.7	0.0	2
	C12H10O	diphenyl oxide	A	57.4	58.7	-1.3	0.0	1.0	1
	C12H10O	diphenyl oxide	A	57.4	58.7	-1.3	0.0	1.0	1
	C12H10O	o-hydroxybiphenyl	B	49.0	50.0	-1.0	0.0	0.0	1
	C12H10O	o-hydroxybiphenyl	A	49.0	50.0	-1.0	0.0	0.0	1
	C12H12	1,8-dimethylnaphthalene	A	46.9	44.2	2.6	0.7	0.0	1
	C12H12	2,3-dimethylnaphthalene	A	51.3	44.2	7.0	0.7	0.0	2
	C12H12	2,6-dimethylnaphthalene	A	65.4	44.2	21.1	0.7	0.0	1
	C12H12	2,7-dimethylnaphthalene	A	63.3	44.2	19.1	0.7	0.0	1
	C12H12N2O	4,4'-diaminodiphenyl ether	D	16.6	58.7	-42.0	0.0	1.0	1
	C12H12N4O	3,3',4,4'-tetraaminodiphenyl ether	A	62.8	58.7	4.1	0.0	1.0	2
	C12H12O4	1,4-dimethylcubane dicarboxylate	B	93.7	71.8	21.9	0.0	2.6	2
	C12H14O4	diethyl o-phthalate	A	66.6	89.2	-22.5	0.0	4.7	1
	C12H15NO2	phenylaminoethylmethacrylate	A	85.6	80.5	5.1	0.0	3.7	2
	C12H16	cyclohexylbenzene	A	54.4	50.0	4.4	0.0	0.0	2
	C12H18	hexamethylbenzene	B	51.7	29.3	22.3	2.5	0.0	1
	C12H19N	carbazole	B	52.2	44.2	8.0	0.7	0.0	1
	C12H22	bicyclohexyl	A	59.1	50.0	9.1	0.0	0.0	2
	C12H22O2	nonyl acrylate	A	98.8	128.4	-29.5	0.0	9.4	2

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C12H22O2	octyl methacrylate	A	104.6	119.7	-15.0	0.0	8.4	2
C12H22O6	oligoethylene butylene glycol adipate	A	112.5	154.5	-41.9	0.0	12.6	2
C12H24	1-dodecene	A	105.1	124.0	-18.9	0.0	8.9	1
C12H24N2O2	n,n'-di-n-propyladipamide	C	79.9	137.1	-57.1	0.0	10.5	1
C12H24O2	butyl octanoate	A	132.8	132.7	0.0	0.0	9.9	2
C12H24O2	dodecanoic acid	B	115.7	132.7	-17.0	0.0	9.9	1
C12H24O2	dodecanoic acid	B	114.5	132.7	-18.2	0.0	9.9	1
* C12H24O2	dodecanoic acid	D	137.4	132.7	4.7	0.0	9.9	1
C12H26	n-dodecane	C	138.8	128.4	10.5	0.0	9.4	1
C12H26	n-dodecane	A	139.7	128.4	11.4	0.0	9.4	1
C12H30O3Si3	1,1,3,3,5,5-hexaethylcyclotrisiloxane		92.5	97.9	-5.4	0.0	5.8	2
C12H30O3Si3	1,1,3,3,5,5-hexaethylcyclotrisiloxane	A	93.6	97.9	-4.3	0.0	5.8	2
C12H30O3Si3	1,1,3,3,5,5-hexaethylcyclotrisiloxane	A	93.7	97.9	-4.2	0.0	5.8	2
C12H30O3Si3	1,1,3,3,5,5-hexaethylcyclotrisiloxane	A	93.5	97.9	-4.4	0.0	5.8	2
C13H9N	7,8-benzoquinoline	A	43.5	50.0	-6.5	0.0	0.0	2
C13H9N	7,8-benzoquinoline	A	43.5	50.0	-6.5	0.0	0.0	2
C13H9N	acridine	A	54.0	44.2	9.7	0.7	0.0	2
C13H9N	acridine	A	54.0	44.2	9.7	0.7	0.0	2
C13H9N	phenanthridine	A	60.2	44.2	16.0	0.7	0.0	2
C13H9N	phenanthridine	A	60.2	44.2	16.0	0.7	0.0	2
C13H10	fluorene	A	50.5	44.2	6.2	0.7	0.0	1
C13H10	fluorene	C	51.3	44.2	7.1	0.7	0.0	1
C13H10N2	diphenylcarbodiimide	A	64.5	67.4	-2.9	0.0	2.1	2
C13H10O	benzophenone	A	56.7	54.4	2.3	0.0	0.5	2
C13H10O	benzophenone	C	55.0	54.4	0.7	0.0	0.5	2
C13H11N	n-methylcarbazole	A	47.3	44.2	3.1	0.7	0.0	2
C13H11N	n-methylcarbazole	A	47.3	44.2	3.1	0.7	0.0	2
C13H12	diphenylmethane	C	63.6	58.7	4.9	0.0	1.0	2
C13H12	diphenylmethane	C	62.3	58.7	3.5	0.0	1.0	1
C13H12N2O	1,3-diphenylurea	A	67.6	71.8	-4.2	0.0	2.6	2
C13H14N2	bis(4-aminophenyl)methane	D	25.4	58.7	-33.3	0.0	1.0	1
C13H19NO2	hexyl phenylcarbamate	B	99.9	111.0	-11.0	0.0	7.3	1
(C13H24O2)n	poly(tridecanolactone)	A	125.0	117.6	7.4	0.0	14.1	2
C13H24O2	tridecanolactone	A	92.6	50.0	42.6	0.0	0.0	1
C13H24O2	tridecanolactone	A	92.6	50.0	42.6	0.0	0.0	2
C13H26	n-heptylcyclohexane	C	95.5	97.9	-2.4	0.0	5.8	1
C13H26O2	tridecanoic acid	B	135.5	141.4	-5.9	0.0	11.0	1
C13H28	n-tridecane	A	136.5	137.1	-0.6	0.0	10.5	1
C13H28O	1-tridecanol	B	147.7	145.8	1.9	0.0	11.5	1

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C13H28O	tri-tert-butylmethanol	B	32.6	40.9	-8.3	1.1	0.0	2
	C14H8O2	anthraquinone	C	59.0	38.5	20.5	1.4	0.0	1
	C14H10	anthracene	C	58.1	38.5	19.6	1.4	0.0	1
	C14H10	anthracene	A	60.1	38.5	21.6	1.4	0.0	1
	C14H10	anthracene	B	59.1	38.5	20.6	1.4	0.0	1
	C14H10	anthracene	C	58.8	38.5	20.4	1.4	0.0	1
	C14H10	phenanthrene	B	48.2	44.2	4.0	0.7	0.0	1
*	C14H10	phenanthrene	C	57.5	44.2	13.3	0.7	0.0	1
	C14H10	phenanthrene	C	46.1	44.2	1.9	0.7	0.0	1
	C14H10	phenanthrene	C	46.1	44.2	1.9	0.7	0.0	1
	C14H10	phenanthrene	B	49.9	44.2	5.7	0.7	0.0	2
	C14H10	phenanthrene	A	44.8	44.2	0.6	0.7	0.0	1
	C14H10O2	benzil	C	61.6	58.7	2.9	0.0	1.0	2
	C14H12	9,10-dihydroanthracene	A	62.4	44.2	18.1	0.7	0.0	2
	C14H12	9,10-dihydrophenanthrene	A	41.7	44.2	-2.5	0.7	0.0	1
	C14H12	trans-stilbene	A	69.7	58.7	11.0	0.0	1.0	2
	C14H12	trans-stilbene	B	68.8	58.7	10.1	0.0	1.0	2
	C14H12O2	diphenylacetic acid	A	74.4	63.1	11.3	0.0	1.6	1
	C14H14	1,2,3,4-tetrahydroanthracene	A	58.9	50.0	8.9	0.0	0.0	2
	C14H14	1,2-diphenylethane	C	70.9	67.4	3.5	0.0	2.1	1
	C14H14	1,2-diphenylethane	A	78.3	67.4	10.9	0.0	2.1	2
*	C14H14	2,2'-dimethylbiphenyl	A	7.8	50.0	-42.2	0.0	0.0	2
*	C14H14	2-ethylbiphenyl	A	7.7	58.7	-50.9	0.0	1.0	2
	C14H14	phenyl-o-tolylmethane	A	68.8	58.7	10.1	0.0	1.0	2
	C14H14N2O3	p-azoxyanisole	B	77.7	67.4	10.3	0.0	2.1	1
	C14H14N2O3	p-azoxyanisole	A	78.9	67.4	11.4	0.0	2.1	2
	C14H18	1,2,3,4,5,6,7,8-octahydroanthracene	A	60.7	50.0	10.7	0.0	0.0	2
	C14H18	1,2,3,4,5,6,7,8-octahydroanthracene	C	51.8	50.0	1.8	0.0	0.0	1
	C14H20	diamantane	A	48.0	32.7	15.3	2.1	0.0	1
	C14H24	perhydrophenanthrene	B	41.8	50.0	-8.2	0.0	0.0	2
	C14H24	perhydrophenanthrene	B	38.4	50.0	-11.6	0.0	0.0	2
	C14H24	perhydrophenanthrene	B	35.6	50.0	-14.3	0.0	0.0	2
	C14H26O2	decyl methacrylate	A	121.9	137.1	-15.2	0.0	10.5	2
	C14H28O	2-tetradecanone	B	160.2	141.4	18.7	0.0	11.0	1
	C14H28O2	tetradecanoic acid	D	114.5	150.1	-35.6	0.0	12.0	1
	C14H28O2	tetradecanoic acid	B	137.8	150.1	-12.3	0.0	12.0	1
	C14H30	n-tetradecane	A	161.5	145.8	15.8	0.0	11.5	1
	C14H30	n-tetradecane	C	154.3	145.8	8.5	0.0	11.5	1
	C14H30O	1-tetradecanol	B	163.1	154.5	8.6	0.0	12.6	1

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FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
C15H10N2O2	4,4'-diphenyl methane diisocyanate	A	87.1	58.7	28.3	0.0	1.0	2
C15H10N2O2	bis(4-isocyanatophenyl)methane	A	87.1	58.7	28.3	0.0	1.0	1
C15H12	4-methylphenanthrene	A	43.5	50.0	-6.6	0.0	0.0	2
C15H16O2	4,4'-dihydroxydiphenyl-2,2-propane	A	69.5	58.7	10.8	0.0	1.0	2
C15H28O2	pentadecanolactone	A	119.1	50.0	69.1	0.0	0.0	1
C15H30	n-decylcyclopentane	A	132.0	124.0	8.0	0.0	8.9	1
C15H30O	2-pentadecanone	B	174.8	150.1	24.7	0.0	12.0	1
C15H30O2	pentadecanoic acid	B	153.0	158.8	-5.9	0.0	13.1	1
C15H32	n-pentadecane	A	156.0	154.5	1.5	0.0	12.6	1
C15H32O	1-pentadecanol	B	172.8	163.2	9.6	0.0	13.6	1
C16H10	fluoranthene	A	48.9	44.2	4.6	0.7	0.0	1
C16H10	pyrene	A	43.4	38.5	4.9	1.4	0.0	1
C16H12Ge	diethynyldiphenylgermane	A	62.8	58.7	4.1	0.0	1.0	2
C16H12Ge	diethynyldiphenylgermane	A	62.8	58.7	4.1	0.0	1.0	2
C16H14O2	1,2-dibenzoylethane	B	93.1	76.1	17.0	0.0	3.1	1
C16H16N2O2	anisaldazine	B	67.3	84.8	-17.5	0.0	4.2	1
C16H22O3Si3	1,1,3,3-tetraethyl-5,5-diphenylcylotrisiloxane	A	65.8	89.2	-23.3	0.0	4.7	2
C16H22O3Si3	diphenyltetramethylcyclotrisiloxane	B	65.7	54.4	11.3	0.0	0.5	1
C16H25NO2	nonyl phenylcarbamate	B	85.8	137.1	-51.2	0.0	10.5	1
C16H32	1-hexadecene	A	108.8	158.8	-50.0	0.0	13.1	1
C16H32	n-decacyclohexane	A	142.2	124.0	18.2	0.0	8.9	1
C16H32O2	hexadecanoic acid	C	163.5	167.6	-4.1	0.0	14.1	1
C16H32O2	hexadecanoic acid	C	163.5	167.6	-4.0	0.0	14.1	1
C16H32O2	hexadecanoic acid	B	160.0	167.6	-7.6	0.0	14.1	1
C16H34	n-hexadecane	A	183.2	163.2	20.0	0.0	13.6	1
C16H34	n-hexadecane	C	177.1	163.2	13.9	0.0	13.6	1
C16H34O	1-hexadecanol	B	181.2	171.9	9.3	0.0	14.7	1
C16H34O	1-hexadecanol (n-cetyl alcohol)	C	108.5	171.9	-63.4	0.0	14.7	2
C16H40O4Si4	octaethylcyclotetrasiloxane	A	122.9	115.3	7.6	0.0	7.9	2
C17H12	1,2-benzofluorene	B	49.3	50.0	-0.7	0.0	0.0	2
C17H12	2,3-benzofluorene	B	47.8	44.2	3.6	0.7	0.0	2
C17H14N2	2,2-bis(4-cyanatophenyl)propane	A	75.0	76.1	-1.1	0.0	3.1	2
C17H14N2	2,2-bis(4-cyanatophenyl)propane	A	75.0	76.1	-1.1	0.0	3.1	2
C17H34O2	heptadecanoic acid	A	176.2	176.3	-0.1	0.0	15.2	1
C17H36	n-heptadecane	A	174.6	171.9	2.7	0.0	14.7	1
C18H12	triphenylene	A	52.5	35.1	17.4	1.8	0.0	1
C18H14	o-terphenyl	A	52.2	44.2	8.0	0.7	0.0	1
C18H14	p-terphenyl	A	72.5	54.4	18.2	0.0	0.5	2
C18H14	p-terphenyl	B	84.4	54.4	30.0	0.0	0.5	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C18H14	p-terphenyl	A	72.5	54.4	18.1	0.0	0.5	2
	C18H15ClSi	triphenylchlorosilane	A	72.5	63.1	9.5	0.0	1.6	2
	C18H15O4P	triphenyl phosphate	A	91.8	89.2	2.6	0.0	4.7	2
	C18H15OP	triphenylphosphine oxide	C	55.5	63.1	-7.6	0.0	1.6	2
	C18H15OP	triphenylphosphine oxide	A	56.1	63.1	-7.0	0.0	1.6	2
	C18H15P	triphenylphosphine	A	55.6	63.1	-7.5	0.0	1.6	2
	C18H18	1-methyl-7-isopropylphenanthrene	C	48.9	54.4	-5.5	0.0	0.5	1
	C18H18ClNS	chlorprothixene	C	75.1	76.1	-1.0	0.0	3.1	2
	C18H18N2O5	ethyl azoxybenzenedicarboxylate	B	53.0	93.5	-40.5	0.0	5.2	1
	C18H21NO	n-(4-methoxybenzylidene)-p-butylaniline	A	111.3	93.5	17.7	0.0	5.2	2
	C18H21NO	n-(4-methoxybenzylidene)-p-butylaniline	A	99.4	93.5	5.9	0.0	5.2	1
	C18H21NO2	n-(2-hydroxy-4-methoxybenzylidene)-p-butylaniline	A	77.6	89.2	-11.5	0.0	4.7	1
	C18H28Si4O4	1,1,3,3,5,5-hexamethyl-7,7-diphenyltetrasiloxane	A	140.1	54.4	85.8	0.0	0.5	2
	C18H30O4	p-diacetylbenzene diethyl ketal	A	79.8	132.7	-52.9	0.0	9.9	2
	C18H30O4	p-diacetylbenzene diethyl ketal	A	79.7	132.7	-53.0	0.0	9.9	2
	C18H30O4	p-diacetylbenzene diethyl ketal	A	79.8	132.7	-52.9	0.0	9.9	2
	C18H36	cyclooctadecane	B	126.8	50.0	76.8	0.0	0.0	2
	C18H36	n-dodecylcyclohexane	C	177.1	141.4	35.7	0.0	11.0	1
	C18H36N2O2	n,n'-di-n-hexyladipamide	C	94.4	189.3	-94.9	0.0	16.8	1
	C18H36O2	octadecanoic acid	C	198.2	185.0	13.3	0.0	16.2	2
	C18H36O2	octadecanoic acid	B	181.2	185.0	-3.8	0.0	16.2	2
*	C18H36O2	octadecanoic acid	B	178.7	185.0	-6.3	0.0	16.2	1
	C18H36O2	octadecanoic acid	C	199.7	185.0	14.8	0.0	16.2	1
	C18H38	n-octadecane	B	203.7	180.6	23.0	0.0	15.7	2
	C18H38	n-octadecane	C	200.7	180.6	20.1	0.0	15.7	1
	C18H38	n-octadecane	A	204.8	180.6	24.2	0.0	15.7	1
	C18H38	n-octadecane	A	201.9	180.6	21.3	0.0	15.7	2
*	C19H16	triphenylmethane	C	49.8	63.1	-13.2	0.0	1.6	1
	C19H16	triphenylmethane	C	57.2	63.1	-5.8	0.0	1.6	1
	C19H16	triphenylmethane	B	60.2	63.1	-2.9	0.0	1.6	1
	C19H20N2O3	4-propionyl-4'-n-butanoyloxyazobenzene	A	80.6	80.5	0.1	0.0	3.7	2
	C19H22ClNO	p-n-hexyloxybenzylideneamino-p'-chlorobenzene	A	95.1	106.6	-11.5	0.0	6.8	2
	C19H22FNO	p-n-hexyloxybenzylideneamino-p'-fluorobenzene	A	70.8	106.6	-35.8	0.0	6.8	2
	C19H22O2	4-methoxy-4'-butoxy-trans-stilbene	B	89.7	102.2	-12.5	0.0	6.3	1
	C19H23NO	n-p-ethoxybenzylidene-p'-butylaniline	B	88.7	102.2	-13.5	0.0	6.3	2
	C19H23NO	n-p-ethoxybenzylidene-p'-butylaniline	A	88.7	102.2	-13.5	0.0	6.3	2
	C19H23NO	p-n-hexyloxybenzylideneaniline	A	98.8	111.0	-12.1	0.0	7.3	2
	C19H38O2	nonadecanoic acid	B	196.0	193.7	2.4	0.0	17.3	1
	C19H40	n-nonadecane	B	202.1	189.3	12.8	0.0	16.8	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C19H40	n-nonadecane	B	196.8	189.3	7.4	0.0	16.8	2
	C20H12	perylene	A	57.9	38.5	19.4	1.4	0.0	1
	C20H14	beta,beta'-binaphthyl	B	84.4	50.0	34.4	0.0	0.0	2
	C20H14	triptycene	A	57.4	35.1	22.3	1.8	0.0	1
	C20H14O4	phenolphthalein	B	150.6	54.4	96.2	0.0	0.5	2
	C21H14N2O4	bis(4-(n-maleicimido)phenyl)methane	D	42.3	67.4	-25.1	0.0	2.1	1
	C20H22N2O	p-n-hexyloxybenzylideneamino-p'-benzonitrile	A	87.8	106.6	-18.8	0.0	6.8	2
	C20H24O2	4-methoxy-4'-pentoxy-trans-stilbene	B	94.6	111.0	-16.3	0.0	7.3	1
	C20H25NO	p-n-hexyloxybenzylidene-p'-toluidine	A	90.8	106.6	-15.8	0.0	6.8	2
	C20H26O2	northindrone	A	82.7	50.0	32.7	0.0	0.0	2
	C20H40O2	eicosanoic acid	B	198.8	202.4	-3.6	0.0	18.3	1
	C20H42	n-eicosane	B	225.5	198.0	27.5	0.0	17.8	2
	C20H42	n-eicosane	B	225.5	198.0	27.5	0.0	17.8	2
	C20H42	n-eicosane	B	229.8	198.0	31.8	0.0	17.8	2
*	C20H42	n-eicosane	C	198.5	198.0	0.5	0.0	17.8	1
	C20H42	n-eicosane	A	219.6	198.0	21.5	0.0	17.8	2
	C21H16	1,2'-dinaphthylmethane	A	82.7	58.7	24.0	0.0	1.0	1
	C21H24N2O3	4-propionyl-4'-n-hexanoyloxyazobenzene	A	99.2	97.9	1.3	0.0	5.8	2
	C21H24Si3O3	cis-tri(methylphenyl)trisiloxane	A	126.6	124.0	2.6	0.0	8.9	2
	C21H24Si3O3	trans-tri(methylphenyl)trisiloxane	A	136.4	124.0	12.4	0.0	8.9	2
	C21H25N	octylcyanobiphenyl	B	87.3	128.4	-41.1	0.0	9.4	2
	C21H25NO5	4'-nitrophenyl-4-n-octyloxybenzoate	B	107.4	137.1	-29.6	0.0	10.5	2
	C21H25NO5	4-nitrophenyl-4'-octyloxybenzoate	B	106.4	137.1	-30.7	0.0	10.5	2
	C21H26O2	4-methoxy-4'-hexoxy-trans-stilbene	B	96.2	119.7	-23.4	0.0	8.4	1
	C21H42O4	1-monostearin	C	170.2	219.8	-49.5	0.0	20.4	1
	C21H44	n-henicosane	B	202.8	206.7	-3.9	0.0	18.9	2
	C22H14O4	1,4-bis(phenylglyoxaloyl)benzene	A	76.0	71.8	4.2	0.0	2.6	2
	C22H18O4	4,4'-dipropanoyloxydiphenyldiacetylene	A	67.8	93.5	-25.7	0.0	5.2	2
	C22H24O6	di(p-methoxyphenyl)-trans-cyclohexane-1,4-dicarboxylate	A	88.2	97.9	-9.7	0.0	5.8	1
	C22H26N2O3	4-propionyl-4'-n-heptanoyloxyazobenzene	A	87.6	106.6	-19.0	0.0	6.8	2
	C22H28O2	4-methoxy-4'-heptoxy-trans-stilbene	B	101.1	111.0	-9.9	0.0	7.3	1
	C22H28O3	northindrone acetate	A	56.9	58.7	-1.8	0.0	1.0	2
	C22H29NO	n-p-n-pentyloxybenzylidene-p'-n-butylaniline	A	75.7	128.4	-52.6	0.0	9.4	2
	C22H44	1,1,10,10-tetramethylcyclooctadecane	B	110.3	50.0	60.3	0.0	0.0	2
	C22H44N2O2	n,n'-di-n-hexylsebacamide	C	129.3	224.2	-94.8	0.0	20.9	1
	C22H46	n-docosane	B	243.5	215.4	28.1	0.0	19.9	2
+	C22H46	n-docosane	B	245.3	215.4	29.9	0.0	19.9	1
	C22H46	n-docosane	B	245.3	215.4	29.9	0.0	19.9	2
	C22H46	n-docosane	B	244.9	215.4	29.5	0.0	19.9	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C23H28N2O3	4-propionyl-4'-n-octanoyloxyazobenzene	A	93.6	115.3	-21.7	0.0	7.9	2
	C23H30O2	4-methoxy-4'-octoxy-trans-stilbene	B	98.9	137.1	-38.1	0.0	10.5	1
	C23H31NO	n-p-n-hexyloxybenzylidene-p'-n-butylaniline	A	76.0	137.1	-61.1	0.0	10.5	2
	C23H44O5	1-aceto-3-stearin	C	130.3	224.2	-93.8	0.0	20.9	1
	C23H48	n-tricosane	B	237.7	224.2	13.5	0.0	20.9	2
	C23H48	tricosane	C	239.9	224.2	15.8	0.0	20.9	2
	C24H18	1,3,5-triphenylbenzene	A	74.9	58.7	16.2	0.0	1.0	2
	C24H18	p-quaterphenyl	B	105.1	58.7	46.4	0.0	1.0	2
	C24H18	p-quaterphenyl	A	64.4	58.7	5.7	0.0	1.0	2
	C24H22O4	4,4'-dibutanoyloxydiphenyldiacetylene	A	85.1	111.0	-25.9	0.0	7.3	2
	C24H24	1,1-dicyclohexyldodecane	B	147.3	145.8	1.5	0.0	11.5	1
	C24H28O2Si3	1,1,1,5,5,5-hexamethyl-3,3-diphenyltrisiloxane	A	84.1	76.1	8.0	0.0	3.1	2
+	C24H30N2O3	4-propionyl-4'-n-nonanoyloxyazobenzene	A	103.9	124.0	-20.1	0.0	8.9	2
	C24H34	1,1-diphenyldodecane	B	148.1	145.8	2.4	0.0	11.5	1
	C24H34	1,1-diphenyldodecane	A	148.1	145.8	2.4	0.0	11.5	2
	C24H40	1-cyclohexyl-1-phenyldodecane	A	127.7	145.8	-18.0	0.0	11.5	2
	C24H40	1-cyclohexyl-1-phenyldodecane	B	127.5	145.8	-18.2	0.0	11.5	1
	C24H46	1,1-dicyclohexyldodecane	A	147.3	145.8	1.5	0.0	11.5	2
	C24H46	2,11-dicyclohexyldodecane	B	147.3	145.8	1.5	0.0	11.5	1
	C24H50	n-tetracosane	B	266.9	232.9	34.0	0.0	22.0	2
	C24H50	n-tetracosane	B	266.9	232.9	34.1	0.0	22.0	2
	C24H50	n-tetracosane	C	253.9	232.9	21.0	0.0	22.0	2
	C25H32N2O3	4-propionyl-4'-n-decanoyloxyazobenzene	A	109.0	132.7	-23.7	0.0	9.9	2
	C25H34O2S	4-n-pentylphenyl-4'-n-heptyloxythiobenzoate	A	101.9	158.8	-56.9	0.0	13.1	1
	C25H34O3	northindrone dimethylpropionate	A	75.6	76.1	-0.5	0.0	3.1	2
	C25H40O2Si2	northindrone pentamethyldisiloxyl ether	A	64.5	63.1	1.5	0.0	1.6	2
	C25H41DO3	p-n-octadecyloxybenzoic acid-d	B	175.6	202.4	-26.8	0.0	18.3	1
	C25H42O3	p-n-octadecyloxybenzoic acid	B	177.3	206.7	-29.3	0.0	18.9	1
	C25H46O6	1,2-diaceto-3-stearin	C	218.7	245.9	-27.2	0.0	23.6	1
	C25H52	n-pentacosane	B	252.4	241.6	10.8	0.0	23.0	2
	C25H52	n-pentacosane	B	258.1	241.6	16.6	0.0	23.0	2
*	C25H52	n-pentacosane	B	243.1	241.6	1.5	0.0	23.0	1
	C26H26O3Si3	dimethyltetraphenylcyclotrisiloxane	B	78.1	63.1	15.0	0.0	1.6	1
	C26H26O4	4,4'-dipentanoyloxydiphenyldiacetylene	A	70.7	128.4	-57.7	0.0	9.4	2
	C26H34N2O3	4-propionyl-4'-n-undecanoyloxyazobenzene	A	119.3	141.4	-22.1	0.0	11.0	2
	C26H38N2O3	4,4'-bis(n-heptyloxy)azoxybenzene	B	82.5	171.9	-89.4	0.0	14.7	1
	C26H52	1,1,4,4,10,10,13,13-octamethyl cyclooctadecane	B	61.8	50.0	11.8	0.0	0.0	2
+	C26H54	n-hexacosane	B	284.5	250.3	34.2	0.0	24.1	1
	C26H54	n-hexacosane	A	287.0	250.3	36.7	0.0	24.1	1

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C26H54	n-hexacosane	B	279.2	250.3	29.0	0.0	24.1	2
	C26H54	n-hexacosane	B	285.3	250.3	35.0	0.0	24.1	2
	C27H30O3	northindrone benzoate	A	78.2	63.1	15.1	0.0	1.6	2
	C27H36N2O3	4-propionyl-4'-n-dodecanoyloxyazobenzene	A	120.8	150.1	-29.3	0.0	12.0	2
	C27H38O2	4-methoxy-4'-dodecoxy-trans-stilbene	B	141.4	171.9	-30.5	0.0	14.7	1
	C27H38O3	northindrone heptanoate	A	63.5	102.2	-38.7	0.0	6.3	2
	C27H48	11-phenyleicosane	C	220.1	211.1	9.0	0.0	19.4	1
	C27H54	11-cyclohexyleicosane	C	180.4	211.1	-30.6	0.0	19.4	1
	C27H56	n-heptacosane	B	272.4	259.0	13.4	0.0	25.1	2
	C28H30O4	4,4'-dihexanoyloxydiphenyldiacetylene	A	123.7	145.8	-22.0	0.0	11.5	2
	C28H32O4Si4	1,1,3,3-tetramethyl-5,5,7,7-tetraphenylcyclotetrasiloxane	B	83.3	63.1	20.2	0.0	1.6	1
	C28H32Si4O4	tetra(methylphenyl)tetrasiloxane	A	114.6	119.7	-5.1	0.0	8.4	2
	C28H38N2O3	4-propionyl-4'-n-tridecanoyloxyazobenzene	A	138.9	158.8	-19.9	0.0	13.1	2
	C28H58	n-octacosane	B	300.3	267.7	32.6	0.0	26.2	2
	C28H58	n-octacosane	B	300.2	267.7	32.5	0.0	26.2	2
	C29H40N2O3	4-propionyl-4'-n-tetradecanoyloxyazobenzene	A	141.8	167.6	-25.7	0.0	14.1	2
	C29H60	n-nonacosane	B	286.0	276.4	9.6	0.0	27.2	2
	C30H22	p-quinquephenyl	A	64.1	63.1	1.1	0.0	1.6	2
	C30H34O4	4,4'-diheptanoyloxydiphenyldiacetylene	A	122.9	163.2	-40.3	0.0	13.6	2
	C30H42N2O3	4-propionyl-4'-n-pentadecanoyloxyazobenzene	A	157.2	176.3	-19.0	0.0	15.2	2
	C30H62	n-triacontane	B	315.0	285.1	29.9	0.0	28.3	2
+	C30H62	n-triacontane	B	310.6	285.1	25.5	0.0	28.3	1
	C31H44N2O3	4-propionyl-4'-n-hexadecanoyloxyazobenzene	A	161.0	185.0	-23.9	0.0	16.2	2
	C31H64	11-n-decylheneicosane	B	252.2	285.1	-32.8	0.0	28.3	1
	C32H38O4	4,4'-dioctanoyloxydinphenyldiacetylene	A	182.4	180.6	1.7	0.0	15.7	2
	C32H39ClO3	northindrone-6-(4-chlorophenyl)-hexanoate	A	69.7	106.6	-36.8	0.0	6.8	2
	C32H46N2O3	4-propionyl-4'-n-heptadecanoyloxyazobenzene	A	175.9	193.7	-17.7	0.0	17.3	2
	C32H66	n-dotriacontane	B	345.1	302.5	42.6	0.0	30.4	2
	C33H34O3	northindrone biphenyl-4-carboxylate	A	68.4	67.4	1.0	0.0	2.1	2
	C33H34O3	northindrone trans-4-hexylcyclohexylcarboxylate	A	56.8	106.6	-49.8	0.0	6.8	2
	C33H40O3	northindrone 4-cyclohexylbenzoate	A	76.4	67.4	8.9	0.0	2.1	2
	C33H48N2O3	4-propionyl-4'-n-octadecanoyloxyazobenzene	A	178.5	202.4	-23.8	0.0	18.3	2
	C33H48O3	northindrone trans-3-(4-butylcyclohexyl)propionate	A	60.2	106.6	-46.4	0.0	6.8	2
	C33H68	n-tritriacontane	B	305.2	311.2	-6.1	0.0	31.4	1
	C34H42O4	4,4'-dinonanoyloxydiphenyldiacetylene	A	143.6	198.0	-54.4	0.0	17.8	2
	C34H70	n-tetratriacontane	B	371.2	319.9	51.3	0.0	32.5	2
+	C34H70	n-tetratriacontane	B	370.0	319.9	50.1	0.0	32.5	1
+	C35H72	n-pentatriacontane	B	366.8	328.6	38.2	0.0	33.5	1
	C36H18	decacyclene	B	80.4	35.1	45.3	1.8	0.0	2

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	FORMULA	NAME	EVAL	ΔS_m^{obs} (J/Kmole)	ΔS_m^{calc} (J/K mole)	DIFF	LN σ	LN ϕ	REF
	C36H24	1,3,5-tri-2-naphthylbenzene	C	89.9	58.7	31.2	0.0	1.0	1
	C36H46O4	4,4'-didecanoyloxydiphenyldiacetylene	A	250.5	215.4	35.1	0.0	19.9	2
	C36H74	n-hexatriacontane	B	371.2	337.3	33.8	0.0	34.6	2
*	C36H74	n-hexatriacontane	B	490.9	337.3	153.6	0.0	34.6	2
	C38H50O4	4,4'-diundecanoyloxydiphenyldiacetylene	A	165.3	232.9	-67.6	0.0	22.0	2
	C39H74	trilaurin	C	386.6	367.8	18.8	0.0	38.2	1
	C40H54O4	4,4'-didodecanoyloxydiphenyldiacetylene	A	244.0	250.3	-6.3	0.0	24.1	2
	C41H72O2	cholesteryl myristate	A	136.7	163.2	-26.5	0.0	13.6	2
	C41H72O2	cholesteryl myristate	B	134.6	163.2	-28.5	0.0	13.6	1
*	C42H66O12	benzene-hexa-n-hexanoate	A	281.7	333.0	-51.3	0.0	34.0	2
*	C42H66O12	benzene-hexa-n-hexanoate	A	281.6	333.0	-51.3	0.0	34.0	1
*	C42H72O12	hexa-o-hexanoyl-scyllo-inositol	A	61.9	333.0	-271.0	0.0	34.0	2
	C43H76O2	cholesteryl palmitate	A	160.4	180.6	-20.2	0.0	15.7	2
*	C45H78O2	cholesteryl oleate	A	9.0	189.3	-180.0	0.0	16.8	2
	C45H80O2	cholesteryl stearate	A	189.9	198.0	-8.1	0.0	17.8	2
	C45H86O6	trimyristin	C	460.9	420.1	40.9	0.0	44.5	1
*	C48H78O12	benzene-hexa-n-heptanoate	A	159.7	385.2	-226.0	0.0	40.3	1
*	C48H78O12	benzene-hexa-n-heptanoate	A	149.6	385.2	-236.0	0.0	40.3	2
*	C54H90O12	benzene hexa-n-octanoate	A	293.8	437.5	-144.0	0.0	46.6	2
*	C54H90O12	benzene-hexa-n-octanoate	A	345.7	437.5	-91.7	0.0	46.6	1
*	C54H96O12	hexa-o-octanoyl-scyllo-inositol	A	124.3	437.5	-313.0	0.0	46.6	2
	C57H110O6	tristearin	C	588.0	524.6	63.4	0.0	57.1	1
	C58H98O6	tripalmitin	C	529.3	472.3	57.0	0.0	50.8	1
*	C66H96O12	2,3,6,7,10,11-hexa-n-octanoyloxy triphenylene	A	108.9	437.5	-329.0	0.0	46.6	2
*	C66H120O12	hexa-o-decanoyl-scyllo-inositol	A	148.6	542.0	-393.0	0.0	59.2	2

* The ΔS_m^{obs} value is either less than 9 J/K·mole or is clearly out of line with other reported values for the same or related compound.

\+ This ΔS_m^{obs} is the value in the original article. It differs from the compiled value.

REFERENCES

1 Domalski, E.S.; Evans, W.H.; Hearing, E.D. Heat Capacities and Entropies of Organic Compounds in the Condensed Phase. *J. Phys. Chem. Ref. Data, 13*, Supplement 1, **1984**, 1-284.

2 Domalski, E.S.; Hearing, E.D. Heat Capacities and Entropies of Organic Compounds in the Condensed Phase, Volume II. *J. Phys. Chem. Ref. Data, 19*, **1990**, 881-1047.